

Corporate Sustainability Data and Machine Learning

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This version: November 2025

Abstract

We first provide a machine learning based framework to estimate large sets of firm sustainability data, which tend to be incomplete but are increasingly important to understand risks related to climate or sustainability oriented policies at the firm or portfolio level. We demonstrate how adapting the framework to different types of data such as carbon emissions or waste-water discharge is simple and does not require domain-specific modeling. This is achieved through using financial data which is abundantly available at the company level. In a second step we quantify the uncertainty of the machine-learning-based point estimates to qualify them for different use cases. Finally, we combine our machine learning framework with recent advances in the field of prediction-powered inference to demonstrate better identification when sustainability labels are scarce and financial data are abundant, the most common regime for firm-level sustainability research and decision making.

Keywords: corporate carbon emissions, corporate sustainability information, machine learning, prediction uncertainty, prediction-powered inference

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We would like to thank Christina E. Bannier, Maurice Dumrose, and Daniela Schoch and the participants in the Irving Fisher Committee on Central Bank Statistics (IFC) and the European Committee on Central Balance-Sheet Data Offices (ECCBSO) Workshop 'New insights from financial statements', the Banking Research Workshop Münster, the International Symposium on Climate, Finance, and Sustainability (ISCFS-2024), and the Sustainable Finance Research Platform Brown Bag Lunch for helpful discussions and comments. The authors acknowledge funding from the project safe Financial Big Data Cluster (safeFBDC) by the Federal Ministry for Economic Affairs, grant number 01MK21002D, and Climate Action and the Stiftung Mercator 'Sustainable Finance Research Platform' (Rahmenprogramm Sustainable Finance, grant number 19026202).

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1. Introduction

Companies, financial institutions, regulators and researchers increasingly require firm level sustainability information in light of international sustainability ambitions (Kunming Declaration, 2021; United Nations, 2015) and growing sustainability risks (Battiston et al., 2017; Emambakhsh et al., 2023; Nieto, 2019)-both to inform decision making and to enable empirical inference related to companies' sustainability. The availability and quality of sustainability data throughout the dimensions environment (E), social (S), and governance (G) are important. However, these data are incomplete across entities and time, often of inadequate granularity and unknown quality, or (artificially) scarce. In this paper, we present a framework that leverages machine learning (ML) to address these issues and fill sustainability data gaps based solely on readily available company data. We include quantification of prediction uncertainty and application of prediction powered inference to allow for a robust use of our framework by decision-makers in the economy and by researchers, respectively.

Companies face regulatory and investor pressures (Slager et al., 2023), customer and employee expectations (Jing et al., 2023; Whelan & Kronthal-Sacco, 2019), and the need to optimize financial performance (Adams & Ferreira, 2009; Kim & Starks, 2016), directing their focus on sustainability across operations and supply chains (Marano et al., 2024). Similarly, financial institutions require data to align portfolios with their sustainability goals (Bolton et al., 2022), cater to the preferences of retail investors for green products (Bauer et al., 2021), or manage financial risks associated with climate and biodiversity impacts (Garel et al., 2023; Giglio et al., 2023; Ilhan et al., 2021, 2023), all within the context of strengthening regulatory frameworks (ECB, 2022). Empirical research on sustainability and business practices likewise depends on high-quality data (Garel et al., 2023; Starks, 2023).

Currently, sustainability at the company level is typically measured by ESG ratings. However, these ratings diverge in terms of the company-level sustainability assessment (Berg et al., 2022), putting into question the usability of these ratings. Given this difficulty, granular physical indicators would enable different actors to assess corporate sustainability more independently. These granular physical indicators include corporate carbon emissions, the corporate biodiversity footprint, corporate resource usage (water, primary materials, etc.), health and safety as well as diversity measures. To serve these data needs, corporate ESG disclosure around the world is currently evolving but far from being established (Krueger et al., 2024). Given the recent

reduction in the scope of European sustainability disclosure regulations under the so-called Omnibus process (European Commission, 2025) and the SECs discontinuation of introducing sustainability disclosure rules in the US (Pinedo et al., 2025), firm sustainability data in major economies are likely to remain scarce for the foreseeable future. Information on supply chains will also remain scarce. This lack of comprehensive data can lead to misinformed decision-making. Therefore, methods are needed to address these data issues.

Various modeling approaches have been proposed to estimate the ESG data. To date, this work has focused mainly on corporate carbon emissions (Busch et al., 2022). Methods include simple estimations, which use a direct proportional relationship between a company's size (e.g., revenue, number of employees) and the corporate carbon footprint, and regression setups, where emissions are regressed against a variety of operational and financial predictors to encapsulate a company's business model, scale, and detailed technological practices (Goldhammer et al., 2017; Griffin et al., 2017). However, existing approaches face challenges: Companies can be complex systems (Loughran & McDonald, 2023). Even within a sector, they are heterogeneous with respect to many aspects that may be important for their physical sustainability metrics. For example, they have different product lines, technologies, market locations, financing structures, performance, or company cultures, which may all play a role for the firm's carbon emissions. First, many of these aspects are not covered by available company-level data. Second, the (statistical) association of accounting data or the relevant aspects above with emissions is not understood and probably highly non-linear. As a consequence, existing approaches suffer from poor out-of-sample predictions and thus remain unreliable as a base for decision making and of limited use for inference frameworks that depend on informative underlying predictions.

ML with its ability to search large non-parametric algorithmic spaces (Jordan & Mitchell, 2015) is therefore well suited to capture the complex mapping from financial and fundamental data to sustainability metrics. There are studies applying ML to corporate sustainability prediction. Although they show encouraging results (Nguyen et al., 2021, 2022), they typically focus on a single metric (carbon emissions), use narrow model classes, and therefore require tremendous metric-specific modeling effort (e.g., Khan et al. (2023), Ranta and Ylinen (2023), and Tian (2023)). The different approaches to applying ML require the user to tailor new data sets, code environments, and algorithms for each sustainability metric. This process is both time-consuming and costly, particularly when a variety of sustainability data points or indicators are needed. From the perspective of asset management or banking, relevant use cases include portfolio decarbonization,

the generation of other portfolio-level sustainability information for reporting purposes, and various forms of sustainability-related risk screening. Additional applications involve firm-level analyses and the identification of environmental or social “hot spots” within supply chains. However, to date, a generalizable framework is lacking that integrates comprehensive coverage across the ESG dimensions.

We propose a general framework that addresses these limitations through two core features: First, it leverages a large but readily available set of firm-level financials and fundamentals, including balance-sheet and profit-and-loss items, as well as structural characteristics, to predict sustainability metrics. Second, the architecture systematically integrates regression model selection and data preprocessing within an end-to-end pipeline, enabling systematic optimization without necessitating domain-knowledge.

Building on this framework, we first test whether sustainability information is captured (and can be recovered) in reported firm-level financial and fundamental data. We apply our framework to five qualitatively different sustainability metrics: Scope 1 and Scope 2 greenhouse gas emissions, air pollution (NO_x emissions), water discharge, and the share of women on the board. We show that our best-performing models compare favorably with existing estimation approaches. We examine how the model performance varies across time, region, and sectors. For the earlier years in the dataset, the predictions tend to be better than in the following years. When looking at different world regions, we also observe variations with the predictions for Europe being more accurate than those in other parts of the world. Across different sectors, the estimation performance of our framework varies and depends on the specific sustainability indicator being predicted.

Second, we extend the framework beyond point estimates and include prediction-uncertainty quantification, arguing that reliable uncertainty measures are essential for many practical and regulatory use cases. We quantify prediction uncertainty using conformalized quantile estimation applied to the best-performing models. The approach provides distribution-aware intervals - on average as well as for each individual prediction - with valid coverage under mild assumptions. In the case of Scope 1 emissions for example, we show that intervals targeted at higher risk tolerance (inner 68% coverage) are typically narrow, whereas intervals targeted at lower risk tolerance (inner 95% coverage) are considerably wider. Comparing these intervals to commonly used performance metrics—such as those based on one or two standard deviations or mean squared error (MSE)—reveals that these heuristics fail to approximate true predictive uncertainty and may be materially misleading.

Third, we evaluate the usefulness of our framework in combination with recent work on semi-parametric inference designed for "sparse-labels, rich-features" settings. Using a simulation study based on real data, we combine the models developed in our framework with prediction powered inference (Angelopoulos et al., 2023, 2024; Zrnic & Candès, 2024) to assess their performance under varying levels of sustainability-data availability. In all settings, the integrated approach shows narrower confidence intervals and improved identification of systematic associations compared to classical estimators based solely on labeled data. These results demonstrate that our framework, when embedded in modern inference procedures, can contribute to enhancing the precision of empirical analysis in environments with limited sustainability data.

Our paper contributes to the literature on sustainability-data by introducing a generalised ML framework that uses widely available financial and fundamental information to produce high-quality firm-level sustainability measures. It connects to recent work in semi-parametric inference, i.a., Angelopoulos et al. (2023), Bates et al. (2024), and Chernozhukov et al. (2018), by showing how the framework can be integrated with modern statistical methods to enable efficient and valid inference on systematic associations under sparse sustainability data. Together, these contributions aim to support both decision-making and growing research on corporate sustainability, i.a., Bolton and Kacperczyk (2021, 2025) and Garel et al. (2025).

The remainder of the paper is structured as follows. Section 2 outlines the general framework, including a description of the data. Sections 3 and 4 present the procedure and results for point-estimation and uncertainty-quantification, respectively. Section 5 demonstrates the relevance of the framework for efficient inference. Section 6 concludes.

2. A general framework for sustainability data estimation

Predicting cardinal sustainability data is a ML regression task. Here, we suggest a versatile framework for using regression-based ML to predict a variety of firm-level sustainability data only from reported financial and fundamental data.

ML and hence our methodology differ substantially from common empirical approaches in business and finance research. Common empirical approaches, especially econometrics, traditionally emphasize hypothesis-driven analysis, seeking to identify causal relationships by imposing strong parametric assumptions and leveraging techniques like instrumental variables or structural modeling. The aim is to understand "why" a

relationship exists and to make policy recommendations. In contrast, ML, and therefore our methodology prioritizes predictive accuracy by exploring complex, non-linear relationships in data through algorithmic flexibility and minimal assumptions about underlying structures. While econometrics excels in causal inference, ML is better suited to generate high-quality predictions, especially in data-rich environments with complex interactions.

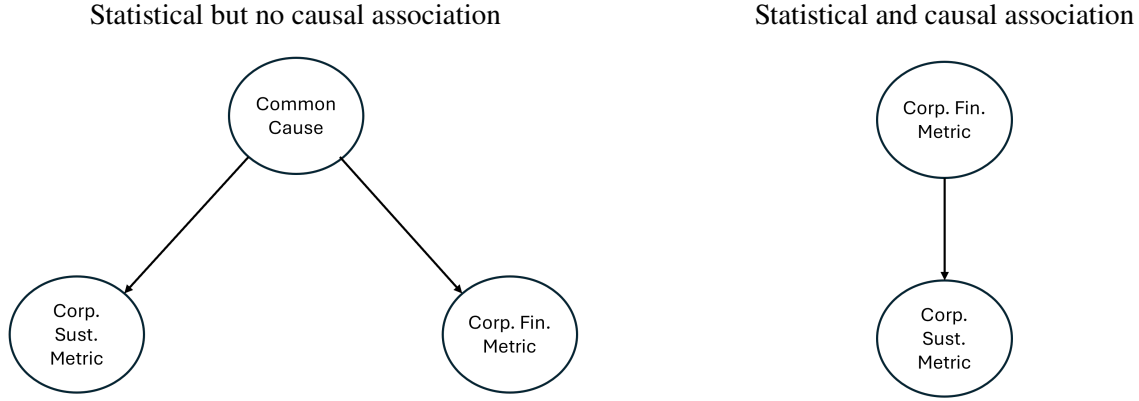


Figure 1. Stylized representations of associations between corporate sustainability and financial metric.

This distinction is visually represented in Figure 1. The left image illustrates a stylized statistical but non-causal association, reflecting the type of complex relationships ML algorithms can effectively exploit for prediction. The right image, on the other hand, depicts an association that is statistical and also causal, which is typically the focus of econometric analysis. Our methodology takes advantage of both types of relationships to maximize prediction performance. As such, the framework is agnostic as to whether a relationship is based on causality or correlation.

2.1 Model space

In our context, a model is defined as a sequence of preprocessing steps followed by a regression algorithm (learner). To ensure the versatility of our framework, we define a large model space that we search using a Bayesian approach¹ to maximize our chance of capturing the "best" model for a given sustainability metric. We maximize the coverage of the model space by first training and optimizing a large set of models (running up to 9,600 trials per sustainability metric²), then optimizing the best-performing model configurations, and

¹Unlike grid or random search, Bayesian optimization utilizes past evaluation results to choose the next set of hyperparameters, efficiently narrowing down to the best possible model settings (Snoek et al., 2012).

²We have devised an early stopping mechanism to boost computational efficiency. For more information, see Appendix E.

finally using them in meta model training, see Figure 2. We evaluate the performance throughout the process using the Mean Squared Error (MSE) and double 10-fold cross-validation.

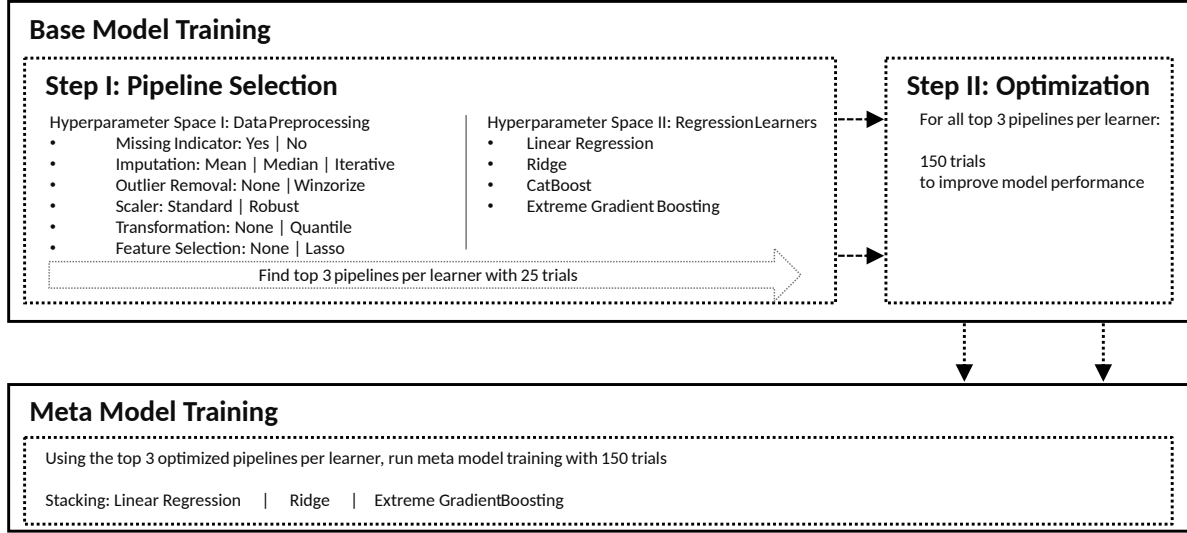


Figure 2. Machine Learning Framework

The figure illustrates the process of Base Model Training, including model selection and optimization using Bayesian hyperparameter tuning, followed by Meta Model Training with a stacked regression approach. Evaluation uses MSE and double 10-fold cross-validation. Base model training involves selecting top pipelines per learner, followed by further optimization. Meta model training uses the top pipelines in different stacking configurations.

Base Model Training - Model Selection: The first step of base model training consists of the search for suitable model configurations for the sustainability metric at hand by exploring a variety of data preprocessing techniques along with different regression learners as input to hyperparameter optimization. In this step, we run up to 25 optimization trials per model configuration.

Data preprocessing is an essential aspect of the modeling process as it can significantly influence the performance of predictive models. In the literature on sustainability data prediction, this step is usually treated separately from the actual model training. Here, we use data preprocessing as a hyperparameter for optimization in itself. In doing so, we expand the considerations of the no free lunch theorem (D. Wolpert & Macready, 1997) to preprocessing in our setup. The options for data preprocessing in our setup span missing indicator flags, imputation methods (mean, median, iterative), outlier removal strategies (none, winsorization), scaling techniques (standard, robust), transformation approaches (none, quantile) and feature selection methods (none, Lasso). We introduce the missing indicator flag (a binary feature) to allow the model to learn from the reporting behavior of a company. The imputation methods are relevant for handling missing data, a common issue in financial datasets. Imputation strategies such as mean and median are simple and widely

used, while iterative methods can provide a more sophisticated approach that utilizes correlations between features (van Buuren, 2007). The removal and scaling of outliers improve the robustness and stability of the models, particularly in financial applications where outliers can represent noise (Aggarwal, 2013). Feature engineering and selection further refine the model by introducing new features that could capture non-linear relationships or selecting the most relevant features to avoid overfitting (Iguyon & Elisseeff, 2003).

In the space of regression learners, the framework contemplates linear regression, ridge regression, extreme gradient boosting (XGBoost) and CatBoost. Linear regression and ridge are fundamental techniques with ridge introducing regularization to manage overfitting (Hoerl & Kennard, 1970). Linear regression is relevant in this study to compare training outputs with methods that are used regularly in econometrics. However, it is unlikely that this learner form the basis of the "best" performing models in the realm of ML. On the other hand, Gradient Boosting algorithms, including XGBoost (Chen & Guestrin, 2016) and CatBoost (Prokhorenkova et al., 2018), are powerful ensemble methods that have shown high performance on a wide range of prediction tasks, particularly in the presence of non-linear and complex relationships.

Compared to Support Vector Machines (SVMs), K-Nearest Neighbors (KNN), neural networks, and simpler tree methods, XGBoost and CatBoost bring a blend of depth and breadth to the modeling process. SVMs, while effective for small to medium-sized datasets, can be outperformed by tree-ensemble methods in handling large and complex data sets (Fernández-Delgado et al., 2014). KNN suffers from the curse of dimensionality and is inherently slower in making predictions due to its instance-based nature (Beyer et al., 1998). Neural networks, although powerful for large-scale and complex non-linear relationships, require extensive tuning and larger datasets to generalize effectively without overfitting (LeCun et al., 2015). This could reduce the versatility of our setup. Simpler tree methods such as CART or C4.5 can provide interpretable models but usually lack the predictive power of boosted ensembles, which aggregate multiple trees to reduce variance and bias (Breiman, 1996). Therefore, for our regression tasks using financial and fundamental data, where feature relationships are likely complex, XGBoost and CatBoost are likely to be good fits.

Base Model Training - Optimization: The second step involves optimizing the top three model configurations per learner and target variable, selected according to their initial performance in Step I. Each model configuration undergoes 150 Bayesian hyperparameter optimization trials. With this step, we substantially expand the optimization efforts to ensure that the model hyperparameters are chosen close to "best".

Meta Model Training: Finally, meta model training is applied using the three best-performing models per learner. The framework employs stacking, which combines the predictions of multiple models by training a meta learner, often leading to performance improvements (D. H. Wolpert, 1992). In this step, we use linear regression, ridge, and XGBoost models but do not repeat the use of CatBoost. In our setting, CatBoost is computationally very costly compared to XGBoost while yielding similar results. The exclusion is thus based on pure efficiency considerations.

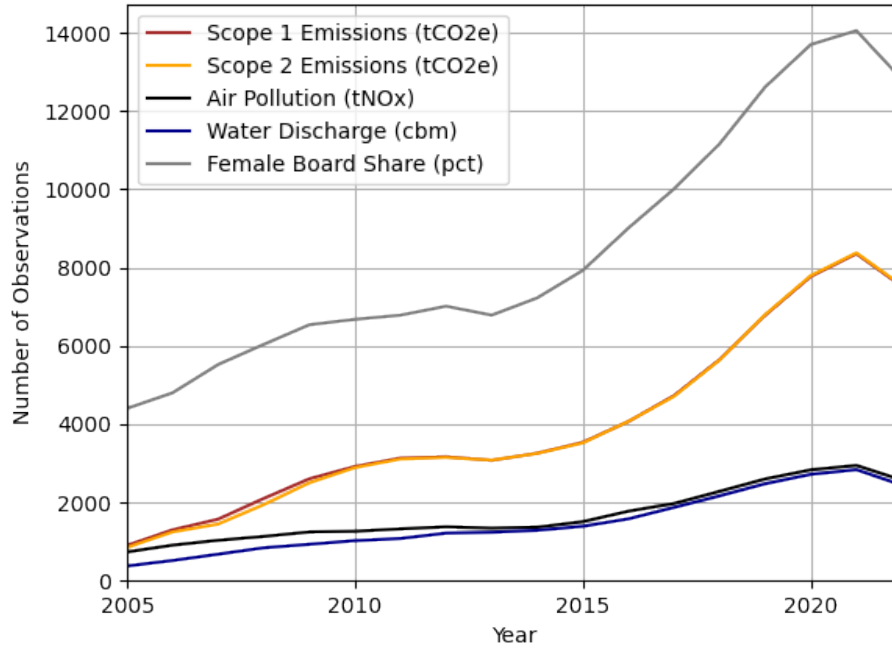
2.2 Training data

Our data set comprises a comprehensive collection of firm-year observations for listed equities obtained from the London Stock Exchange Group (LSEG) Data and Analytics database. This data set is global in scope, encompassing data from 95 countries, which provides a diverse and extensive foundation for modeling and analysis. For this study, only reported sustainability data are included. This avoids potential biases introduced by the LSEG Data and Analytics' estimation models or unaudited data. Reported data may, however, in itself reflect a bias, e.g., due to larger or cleaner firms being more likely to report. The former is considered in Appendix B; the latter cannot be excluded, although most listed firms can be expected to report on emissions only if they are obliged to do so.

As shown in Figure 3, the data set includes key sustainability indicators such as Scope 1 and Scope 2 emissions, air pollution, water discharge, and share of female board members (our "target variables"). Data availability has recently increased substantially, enabling ML applications in the field.

Table I provides a detailed summary of the data, highlighting the breadth and depth of the data set. The data set time period runs from 2005 to 2022.

- **Scope 1 and Scope 2 Emissions:** The dataset contains nearly 50,000 observations each for these emission categories, covering 83 sectors across 83 countries, with data available for more than 8,000 companies. We select these indicators to benchmark against other studies and due to the relevance of climate change to business.
- **Air Pollution and Water Discharge:** These variables have fewer observations, reflecting the more limited availability of environmental data. However, they still provide significant coverage, with data on more than 3,000 companies from more than 60 countries. These indicators are included to reflect



The figure illustrates the availability of reported sustainability data by listed companies over time. The decline in number of observations in the last year of our period is very likely due to a time lag in LSEG data management.

Figure 3. Reported Sustainability Data over Time

emerging topics in business and finance research, such as biodiversity, blue economy, and livable cities.

- **Female Board Share:** This social governance indicator is well-represented, with more than 100,000 observations across 95 countries and 86 sectors, providing the richest data set in our analysis. It should demonstrate that our framework is applicable beyond environmental sustainability data.

The data set includes 211-255 predictor variables, depending on the specific sustainability metric, with data completeness ranging from 57% to 65%. This breadth of variables offers a comprehensive view of company characteristics and operational contexts.

The features selected for this study focus on fundamental and financial data, excluding direct sustainability-related metrics (e.g., energy use) to challenge the model's ability to infer sustainability data solely from financial and fundamental data. The feature set includes financial data, fundamental data, and temporal effects. Financial data (e.g., income statement, balance sheet, cash flow metrics) capture idiosyncratic characteristics of companies such as size, profitability, innovative capacity, and asset intensity. Fundamental data

Dataset	Scope 1 Emissions	Scope 2 Emissions	Air Pollution	Water Discharge	Female Board Share
General Information					
Number of observations	47685	47320	20980	18426	108834
Number of sectors	83	83	76	74	86
Number of countries	83	83	63	63	95
Number of companies	8391	8335	3389	3098	14406
Start year	2005	2005	2005	2005	2005
End year	2022	2022	2022	2022	2022
Number of predictor variables	240	240	213	211	255
Data completeness (in %)	63.86	63.77	65.72	65.62	57.92
Target Variable Information					
Mean	3786965	1051686	19914	187082423	16.00
Standard deviation	26440465	48614503	179304	1318742226	14.00
Minimum	0.00	0.00	0.00	0.00	0.00
Maximum	4421000000	7386660000	14042000	26877900000	100.00
Target Variable Information					
Log (1+value) Mean	10.75	11.07	6.31	14.98	2.17
Log (1+value) Std	3.55	2.72	3.28	3.60	1.40
Log (1+value) Min	0.00	0.00	0.00	0.00	0.00
Log (1+value) Max	22.21	22.72	16.46	24.01	4.62

This table presents summary statistics for the dataset, including Scope 1 and Scope 2 emissions, air pollution, water discharge, and female board share. The table details general information such as the number of observations, sectors, countries, companies, the time period (2005-2022), and data completeness rates. It also includes target variable information, such as means, standard deviations, and the range (minimum to maximum) of the absolute and log-transformed values.

Table I. Summary Statistics

include industry indicators to account for general trends such as emission intensity within specific sectors, spatial variables that reflect policy and socio-economic operating conditions, and company age that reflects the general development stage of a company. Temporal effects are captured by including year variables. A list of all variables including summary statistics is presented in Appendix I.

Data preprocessing steps included retrieving data via the LSEG Data and Analytics API, filtering out non-listed equity observations, and excluding entries without date variables or those with quarterly reporting. The target variables were logarithmically transformed to improve predictive performance. For highly correlated features (correlation $\geq 99\%$) only one feature is retained based on the highest number of observations to mitigate high multicollinearity. Furthermore, features with very low data availability (missingness $\geq 99\%$) were excluded, as imputation was not feasible in these cases.

This rigorous data selection and processing framework prior to the pipeline preprocessing steps ensures that the resulting model is robust and reliable, providing information on how financial metrics can be used to predict sustainability data.

3. Point prediction of sustainability metrics

3.1 Model performance metrics

We evaluate the performance of the top three models from both base and all meta-training using performance assessment metrics as summarized in Table II. We evaluate the point estimates using these metrics.

Metric	Measure/Value	Description
<i>Global Assessment</i>		
Accuracy	Mean Absolute Error (MAE)	The average of the absolute errors between the predicted and actual values, representing average prediction error.
	Mean Squared Error (MSE)	The average of the squared errors, emphasizing larger errors more than MAE. Also used in the model training as loss function.
	R^2 (R-squared)	Measures the proportion of the variance in the dependent variable that is predictable from the independent variables.
	Quintile Deviation	The absolute difference in quintiles between the predicted quintile and the actual sample quintile.
Error Distribution	Graphical representation	Refers to the distribution of relative errors between predictions and actual values.
<i>Local Assessment (Context)</i>		
Sectoral Accuracy	Within-Sector MAE/MSE	Evaluation of model performance within different industry sectors.
Temporal Accuracy	Within-Year MAE/MSE	Evaluation of model performance over different time periods.
Spatial Accuracy	Within-Country MAE/MSE	Evaluation of model performance across different geographical locations.

This table reports a summary of the performance metrics used to assess the point estimates of trained models. Global and local in this context do refer to full or partial results of the model, not to geographic coverage.

Table II. Performance Assessment Metrics for Machine Learning Models

Global assessments encompass a set of quantitative and graphical measures designed to evaluate the prediction performance of global models. The metrics in this category include the accuracy and the error distribution between the predicted and actual values. Accuracy measures include MAE and MSE, which quantify the average prediction error and the average of squared errors, respectively, with MSE placing greater emphasis on larger errors. We use MSE as the metric in our loss function and thus as our main measure of model performance. The R-squared (R^2) measures the proportion of variance in the dependent variable that the model can capture, providing insight into the explanatory power of the model.

Local assessments show how the models perform in their operational contexts. This category includes quintile allocation, which evaluates the model’s ability to accurately rank predictions within specific quintile brackets. This could be useful when seeking best-in-class investment strategies or when reducing exposure

to highly emitting companies in portfolios. The context examines the performance of the model in sector, space, and time. This should ensure transparency about performance differences prevalent in other sustainability data sets (Dobrick et al., 2023).

3.2 Global model assessment

Our analysis reveals that sustainability data can be predicted from corporate financial data with a high degree of accuracy³. Complex and non-parametric ML models significantly outperform linear models in predicting sustainability data from financial data, see Table III. This indicates that the underlying structure of sustainability data is quite complex and cannot be adequately captured by simple linear relationships.

Target Variable	Metric	Base Learner				Meta Learner		
		Linear Regression	Ridge	CatBoost	XGBoost	Linear Regression	Ridge	XGBoost
Scope 1 Emissions (tCO ₂ e)	MAE	1.261	1.240	0.648	0.662	0.610	0.609	0.589
	MSE	3.401	3.252	1.579	1.617	1.622	1.622	1.589
	R2	0.713	0.726	0.867	0.864	0.863	0.863	0.866
Scope 2 Emissions (tCO ₂ e)	MAE	1.252	1.118	0.513	0.627	0.556	0.556	0.527
	MSE	3.400	2.959	1.163	1.269	1.275	1.275	1.204
	R2	0.540	0.599	0.842	0.828	0.827	0.827	0.837
Air Pollution (tNO _x)	MAE	1.501	1.444	0.695	0.905	0.672	0.672	0.650
	MSE	4.598	4.320	1.923	2.262	1.907	1.907	1.909
	R2	0.564	0.590	0.818	0.786	0.819	0.819	0.819
Water Discharge (cbm)	MAE	1.782	1.782	0.278	0.623	0.275	0.275	0.273
	MSE	7.528	7.522	1.055	1.526	1.063	1.063	1.053
	R2	0.443	0.443	0.922	0.887	0.921	0.921	0.922
Female Board Share (pct)	MAE	0.876	0.876	0.473	0.501	0.448	0.448	0.428
	MSE	1.253	1.249	0.598	0.638	0.595	0.595	0.592
	R2	0.347	0.349	0.688	0.667	0.690	0.690	0.691

This table reports the performance metrics of the ML models Linear Regression, Ridge, CatBoost, and XGBoost, used to predict sustainability data. The models are evaluated based on their MAE, MSE, and R-squared (R2) values in five target variables Scope 1 Emissions, Scope 2 Emissions, Air Pollution, Water Discharge, and Female Board Share. Target variable is in log + 1 format. The results are shown for the best performing pipeline, that is, the pipeline that "won" the horse race. In Appendix A, we provide a visual representation of the global results.

Table III. Global Model Performance

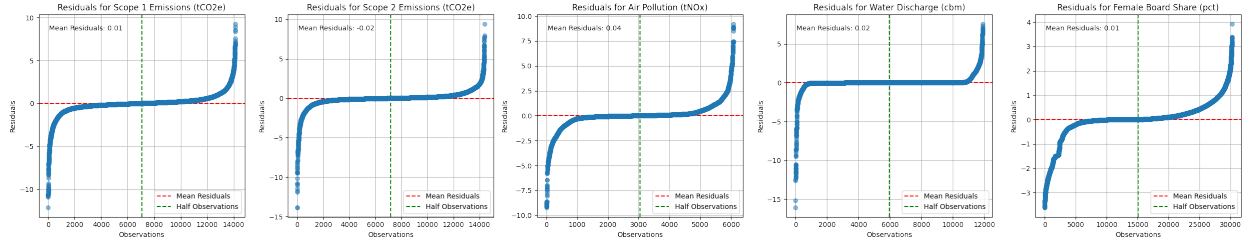
Specifically, our best-performing models achieve MSE in the range of 0.6-1.9 and MAE between 0.3 and 0.7. These results within our data environment are substantially better compared to a benchmark study on corporate carbon emissions by Nguyen et al. (2022), which reported MAE values around 1.1 and 0.8 for Scope 1 and Scope 2 emissions where we achieve 0.6 and 0.5 respectively. Moreover, our models demonstrate high explanatory power. With the exception of the female board share, the best performing models explain between 82 and 92% of the variance in the sustainability data sample. This high level of performance underscores the

³For further information on the computational efficiency of our framework, see Appendix F

potential of advanced ML techniques in enhancing the predictive accuracy of sustainability metrics based on financial data.

For better readability, we show the results for the best-performing model for each of the sustainability metrics in subsequent analyzes. These are base XGBoost models for Scope 1 and 2 emissions, a meta Ridge model for air pollution, and meta XGBoost models for water discharge and female board share.

The global models might bias towards over- or underestimating the target variables structurally. Figure 4 shows, however, that the residuals of the best performing models are evenly distributed between over- and underperformance in number of observations and in magnitude of the residuals. This is also reflected in the near-zero mean of the residuals. Only for female board share, we do not find fully symmetrical residuals. Hence, our models perform well for all target variables, especially for the environmental ones.



This figure shows the ranked distribution of residuals in the test dataset for five different variables: Scope 1 emissions, Scope 2 emissions, air pollution, water discharge, and female board share.

Figure 4. Ranked distribution of residuals

3.3 Local model assessment

The relative errors in the predictions remain non-negligible. To show the applicability of the predicted data to different use cases, we dissect the results at the local level in the next step.

Quintile Analysis

First, we evaluate the performance of our models using a quintile-based approach. Specifically, we divide the data set into quintiles with roughly the same number of observations based on the size of the target variable⁴. This allows us to assess how well our models perform across different ranges of sustainability metrics.

The quintile-based approach shows that the models generally perform the worst in the smallest quintile, which includes the lowest values of the target variables; see Table IV. This pattern is consistent in the five

⁴Note that female board share has nearly 30% of observations equaling zero. Therefore, in this case, we manually adjusted the binning, so that more observations are in Quintile 1 than in the remaining four quintiles in this case.

target variables analyzed. A key factor contributing to this performance disparity is the inherent difficulty in predicting values close to zero, which is only necessary in the smallest quintile.

Target Variable	Metric	Q1	Q2	Q3	Q4	Q5
Scope 1 Emissions (tCO2e)	MAE	1.112	0.571	0.476	0.466	0.618
	MSE	3.922	0.903	0.718	0.757	1.593
Scope 2 Emissions (tCO2e)	MAE	1.004	0.377	0.404	0.335	0.446
	MSE	3.517	0.413	0.500	0.405	0.982
Air Pollution (tNOx)	MAE	1.137	0.464	0.451	0.698	0.612
	MSE	4.074	0.780	0.803	2.015	1.861
Water Discharge (cbm)	MAE	0.532	0.222	0.205	0.185	0.221
	MSE	3.146	0.467	0.482	0.362	0.804
Female Board Share (pct)	MAE	0.692	0.301	0.315	0.344	0.401
	MSE	1.376	0.254	0.293	0.331	0.448

This table presents the performance metrics of the best-performing ML model across different quintiles of the target variables. The dataset is divided into quintiles based on the size of each sustainability metric Scope 1 Emissions, Scope 2 Emissions, Air Pollution, Water Discharge, and Female Board Share. Performance is measured using MAE and MSE for each quintile, labeled Q1 through Q5, where Quintile 1 (Q1) represents the smallest values of the target variable, while Quintile 5 (Q5) represents the largest values. All input values are in log + 1 format. A visual representation of correct quintile match is available in Appendix C.

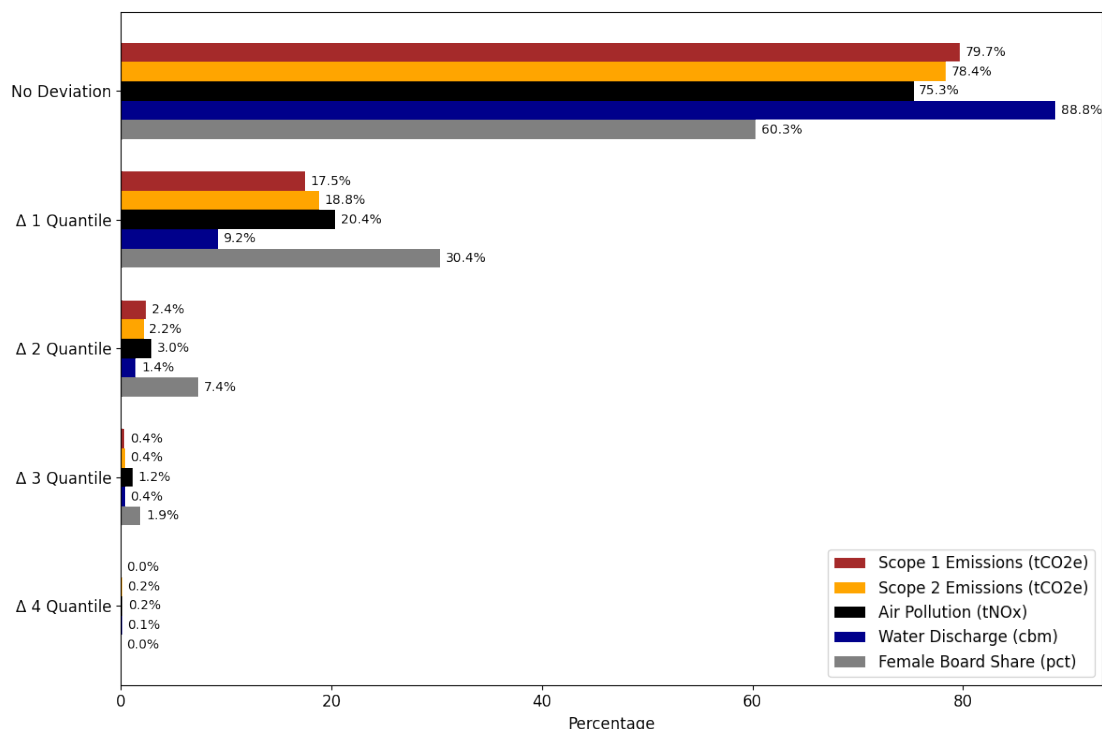
Table IV. Performance of Best Models by Quintiles

The performance variation between the four remaining quintiles is relatively modest, with some target variables showing a slight decrease in performance in the fifth (largest) quintile. We do not observe any particular quintile consistently performing the best or second worst across all metrics, and the deviations in performance are generally small. This suggests that the models maintain a stable performance level in most of the predicted data distributions.

Finally, we categorically examine the degree of deviation between the predicted and actual quintiles. We find that our predictions fall into the right quintile 60.3% (female board share) to 88.8% (Scope 1 emissions) of the time, see Figure 5. In most cases of deviation, the predicted quintile deviates by only one quintile from the actual value. This underscores the overall robustness and performance of our models for categorizing companies based on our continuous predictions.

In practical terms, for a company, these findings indicate that the models are reliable for identifying areas of high-impact within supply chains. This can be particularly useful for companies seeking to prioritize their strategic focus and allocate resources effectively to areas with the greatest impact on sustainability. Similarly, financial institutions can perform heat mapping or clustering exercises in their portfolios using the data.

Next, we analyze the performance of the model in the temporal, spatial, and sectoral dimensions to assess



The bar charts display the proportion of observations of absolute deviating quintiles for the key variables: Scope 1 and Scope 2 emissions, air pollution, water discharge, and female board share.

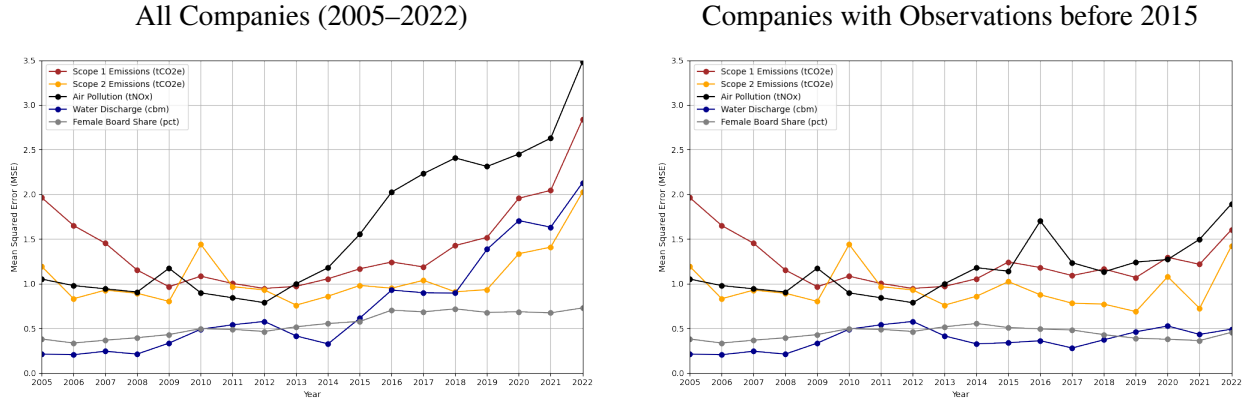
Figure 5. Number of Deviating Quintiles

how our models perform under different conditions.

Temporal Dimension

Our analysis indicates that the models perform best during the period up to 2015, with a slight decrease in performance in subsequent years; see the left panel Figure 6. This decline is relatively minor for all target variables. We attribute it to the increased variability in the data set over time, that is, more companies reporting their sustainability data. Although more data should generally help in training ML models, higher heterogeneity among reporting companies (business models, sizes, technology mix, etc.) make it more difficult for the algorithm to predict with the same performance level, especially if the heterogeneity grows faster than the data availability. The flattened curves in the right panel of Figure 6 are consistent with this conjecture. It shows the same plots but for companies for which data are already available before 2015. We deduce from the comparison of the full set and the subset of the data that the heterogeneity of companies that started to report relatively recently drives the deterioration in prediction performance over time.

When applying ML models to real-world scenarios, it is essential to consider these temporal variations. Users



This figure illustrates the model performance over time (2005–2022) for five different variables: Scope 1 emissions, Scope 2 emissions, air pollution, water discharge, and female board share. The left panel shows results for all companies, while the right panel includes only companies with observations recorded before 2015. The ordinate represents the MSE, showing how prediction performance has evolved for each variable across the years.

Figure 6. Temporal Model Performance

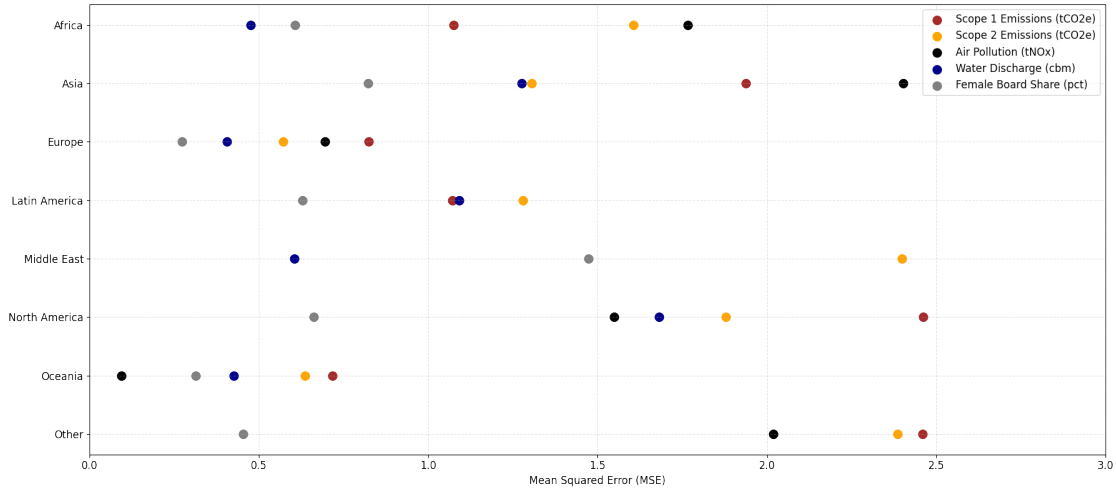
should ensure that the specific use case is well understood so that the models can be appropriately adapted. Particularly when focusing on current data, it might be necessary to apply variations or modifications to the models trained on older data. One potential approach is to incorporate a discount factor in the loss function that adjusts for temporal discrepancies.

Spatial Dimension

In the spatial analysis, we observe some variation in model performance between different geographic regions; see Figure 7. In particular, our models tend to perform better in Europe (and Oceania) than in other parts of the world. This discrepancy could be partially attributed to stricter and more comprehensive reporting regimes in Europe, resulting in a larger and more reliable dataset (Krueger et al., 2024), that is, the reporting is more homogeneous due to mandatory guidelines that improve model performance in these regions (or reduce heterogeneity). However, despite these differences, the models still perform well in all regions. For companies looking to optimize supply chains across world regions, our framework is therefore applicable, although caution is advised.

Sectoral Dimension

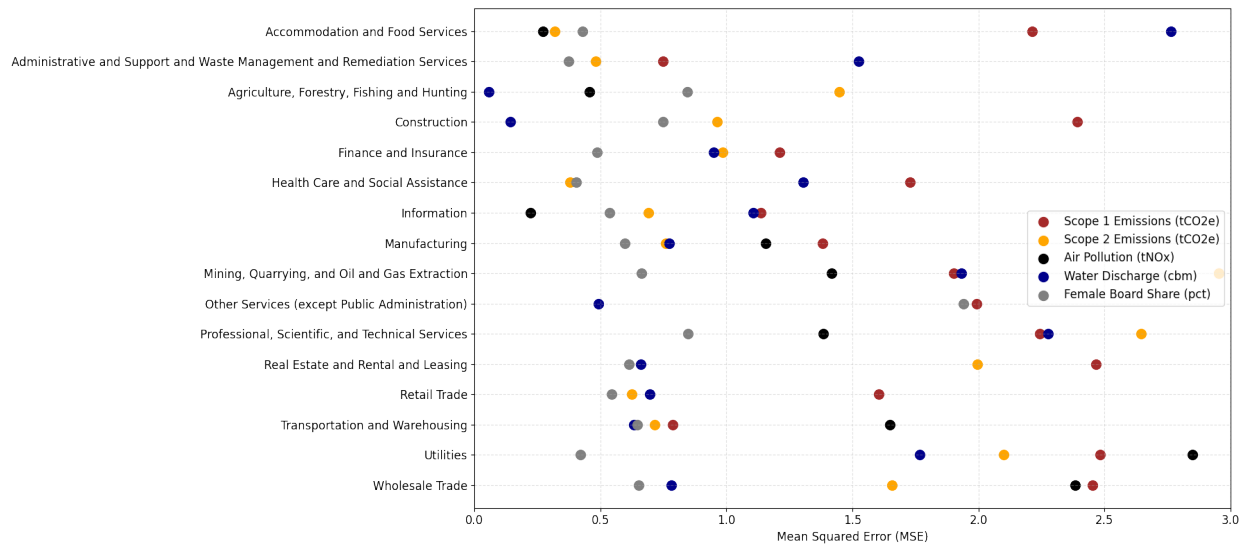
Regarding the sectoral dimension, we find minor variations in model performance across different industries, with some outliers; see Figure 8. For most industries, the models perform reliably, suggesting that they are well suited for applications that involve analyzing portfolios spanning multiple sectors and industries, such



This figure shows the model performance across different global regions, as measured by the MSE for the five target variables: Scope 1 emissions, Scope 2 emissions, air pollution, water discharge, and female board share. Each region is represented along the ordinate, while the MSE values are plotted as dots.

Figure 7. Spatial Model Performance

as those of banks or asset managers. However, if a particular sector is of special importance to an economic actor, it may be beneficial to use a weighted loss function tailored to sectoral importance.



This figure displays the MSE of model predictions across industries for the five target variables: Scope 1 emissions, Scope 2 emissions, air pollution, water discharge, and female board share. Each dot represents the MSE for a particular variable within a specific industry.

Figure 8. Sectoral Model Performance

In summary, while there is some variation in the model performance across the temporal, spatial, and sectoral dimensions, the models generally perform well in all three areas. However, the variation underscores the importance of the model-flexibility within our approach to be best suited for the task at hand. In Appendix

G, we provide some recommendations for researchers on how to work with our framework and the predicted sustainability data given these variations.

4. Quantifying prediction uncertainty

4.1 Approach

Point estimates generated by classical statistical or ML models represent the conditional expectation of the target variable - in our case the conditional mean - but do not provide information about the uncertainty associated with the predicted value. However, quantifying this uncertainty is crucial to make reliable statements about the accuracy of the estimation, to possibly exclude certain predictions from use (*measure not estimate*), or to opt for more conservative or optimistic values instead of the point estimates of the conditional mean.

A key challenge in accurately and consistently assessing the uncertainty of different models is the construction of valid prediction intervals. Our selection of approaches and methods for generating prediction intervals is, therefore, guided by the goal of achieving (sufficiently) accurate marginal coverage and, ideally, conditional coverage.

Marginal coverage refers to the probability that, on average across all predictions, the actual value of the target variable lies within the prediction interval. This property ensures that the prediction intervals are correct on average across the entire distribution of input data. Marginal coverage does not guarantee accurate coverage for every individual prediction, but does ensure that the average coverage meets the desired level, e.g. to generate aggregate portfolio metrics.

Conditional coverage refers to the probability that the true value of the target variable lies within the predicted interval for a specific set of input values. Achieving conditional coverage requires that the prediction intervals are accurately calibrated for each possible input, capturing the uncertainty for that particular instance. This property is stronger than marginal coverage and is crucial when individual-level uncertainty quantification is needed, such as in company-specific predictions, e.g. to identify outliers or hot spots.

In our data-driven framework, where financial data are used as predictors of sustainability-related information, additional challenges arise due to the unknown relationship between these variables (absence of theory). This suggests a preference for methods that do not rely on strong assumptions about the joint distribution.

In the literature on uncertainty quantification, there are various approaches to quantifying uncertainties related to model prediction (Abdar et al., 2021; Soize, 2017). A common method involves using scalar uncertainty measures, such as the standard deviation (error), which provides a general measure of the spread of predictions. However, these methods offer only a global perspective on the underlying uncertainty and typically fail to adequately account for the variability of uncertainty across different areas of the distribution. Moreover, theoretical guarantees regarding marginal coverage and conditional coverage are only available under strong assumptions about the distribution and are typically not empirically validated in finite samples (Palmer et al., 2022).

Quantile regression (Koenker & Bassett, 1978) offers an alternative approach that allows the calculation of prediction intervals using the estimation of quantiles instead of the conditional mean of the target variable. This method is particularly useful when asymmetric uncertainties in predictions cannot be ruled out. A central advantage of quantile regression is that it asymptotically ensures both marginal coverage and conditional coverage for sufficiently large sample sizes (Chernozhukov et al., 2009; Romano et al., 2019). In Appendix H, we provide a short demonstration of the properties of quantile regressions.

A relatively new framework, useful for the quantification of prediction uncertainty is conformal prediction (Lei & Wasserman, 2014; Lei et al., 2018; Vovk et al., 2005). The underlying methods provide theoretical guarantees for marginal coverage in finite samples without making strong assumptions about the distribution.⁵

We employ a conformalized version of the quantile regression (Romano et al., 2019). This approach retains the favorable properties of quantile regression concerning the adaptivity of the prediction intervals and (asymptotically) conditional coverage, while also providing theoretical guarantees for marginal coverage in finite samples under the assumption of independent and identically distributed (i.i.d.) data. As with our systematic framework to identifying the optimal model for predicting the conditional mean, this approach to uncertainty quantification aims to be sufficiently close to optimal to allow meaningful conclusions.

Specifically, our framework involves generating symmetric prediction intervals with a target coverage rate of $\tau \in \{68\%, 95\%\}$ to assess prediction uncertainty. Given the size of the datasets, we employ a split method in which the test data set used for point estimates is divided into a calibration set and an uncertainty test set (on top of the training set). This approach is intended to balance the trade-off between statistical efficiency

⁵Our framework builds on work that assumes exchangeability, which is satisfied under the assumption of i.i.d. data. For conformalized prediction approaches beyond exchangeability, see Tibshirani et al. (2019) and Barber et al. (2023).

and computational efficiency. The training of the models using an adapted loss function is performed on the training set. The calibration set is then used to compute conformal scores, which are subsequently used to construct prediction intervals that achieve the desired coverage levels. The test set is used to evaluate the coverage and analyze prediction uncertainty. The implementation steps of this approach are documented in Appendix H.

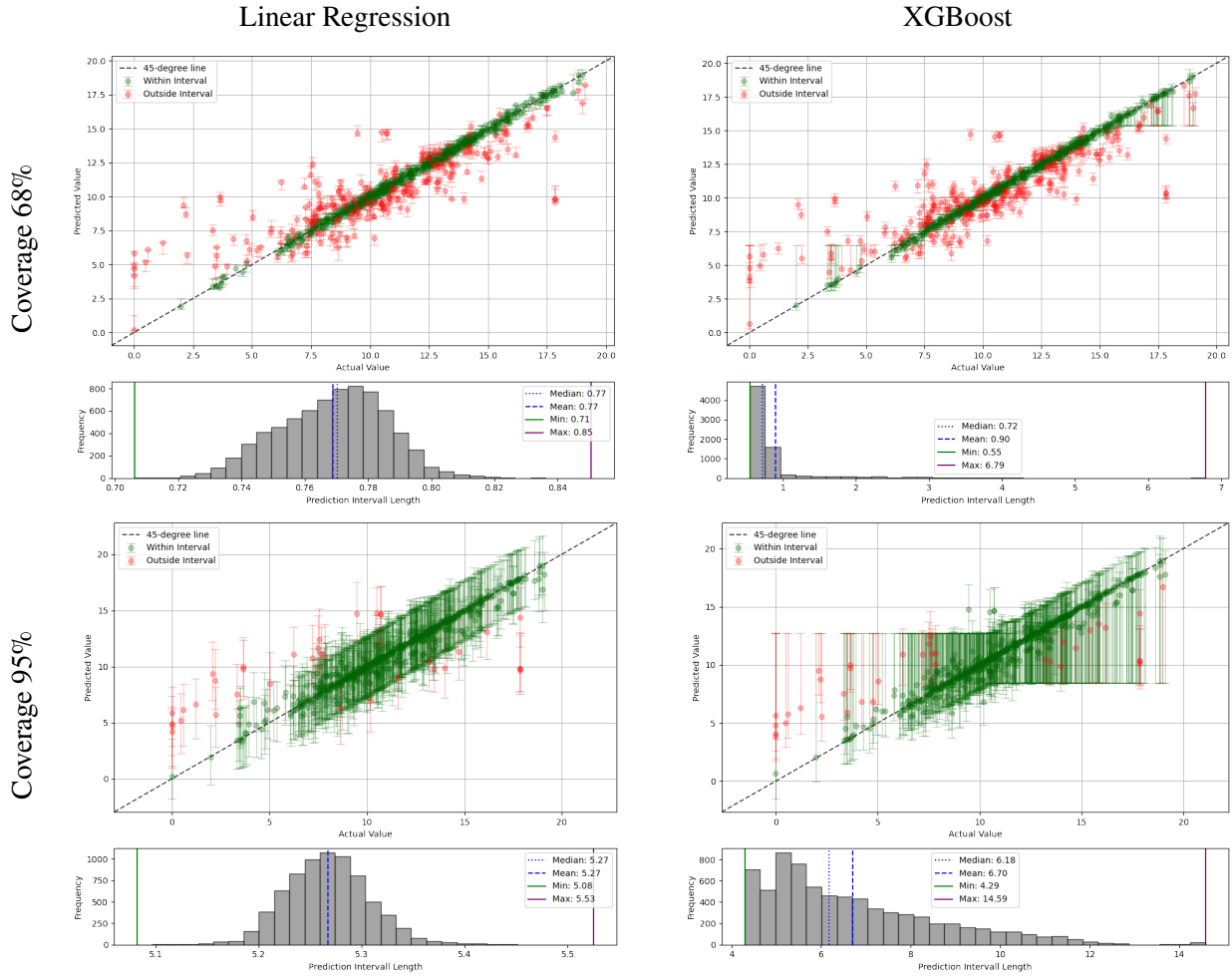
4.2 Prediction uncertainty measure

The analysis of prediction uncertainty across the target variables reveals heterogeneous patterns. For smaller coverage requirements, the uncertainty is generally negligible, whereas for larger coverage requirements, it becomes significant. Our conformalized quantile regression approach demonstrates empirically valid coverage (with an absolute difference between empirical and target coverage $\leq 0.68\%$ for all variables and coverage rates). The prediction uncertainty remains largely symmetrical, balancing under- or overestimation of prediction uncertainty compared to other standard uncertainty measures. In this section, we show the results for Scope 1 Emissions; see Appendix D for the results for the other target variables.

Figure 9 illustrates the prediction uncertainty in four settings, including conformalized quantile regression using linear regression and XGBoost and with coverage rates of 68% and 95%. The figure demonstrates that, while prediction uncertainty always exists, it is relatively low for the 68% coverage rate. This suggests that users who are not highly risk-averse can use the point estimates without encountering (on average) significant uncertainty. However, at the 95% coverage rate, uncertainty increases substantially, indicating a wider range of possible outcomes, which may be of concern to users who require greater confidence in their predictions.

In addition, the choice of the learner to estimate the uncertainty affects the results. Linear regression models tend to produce on average narrower prediction intervals with relatively constant interval lengths across the distribution of the target variable. In contrast, XGBoost exhibits a more adaptive behavior, with wider prediction intervals towards the extremes of the target variable values. For lower actual values, the uncertainty intervals become larger, particularly skewing toward values above the actuals. Similarly, for the largest values, the intervals widen, though skewed downward, toward lower-than-actual values.

Figure 10 further examines the distribution of the prediction intervals, confirming their general symmetries. In this figure, we use predictions of the conditional mean from the (best performing) XGBoost meta-learner. Linear regression models show tighter distributions of interval lengths, while XGBoost produces more widely

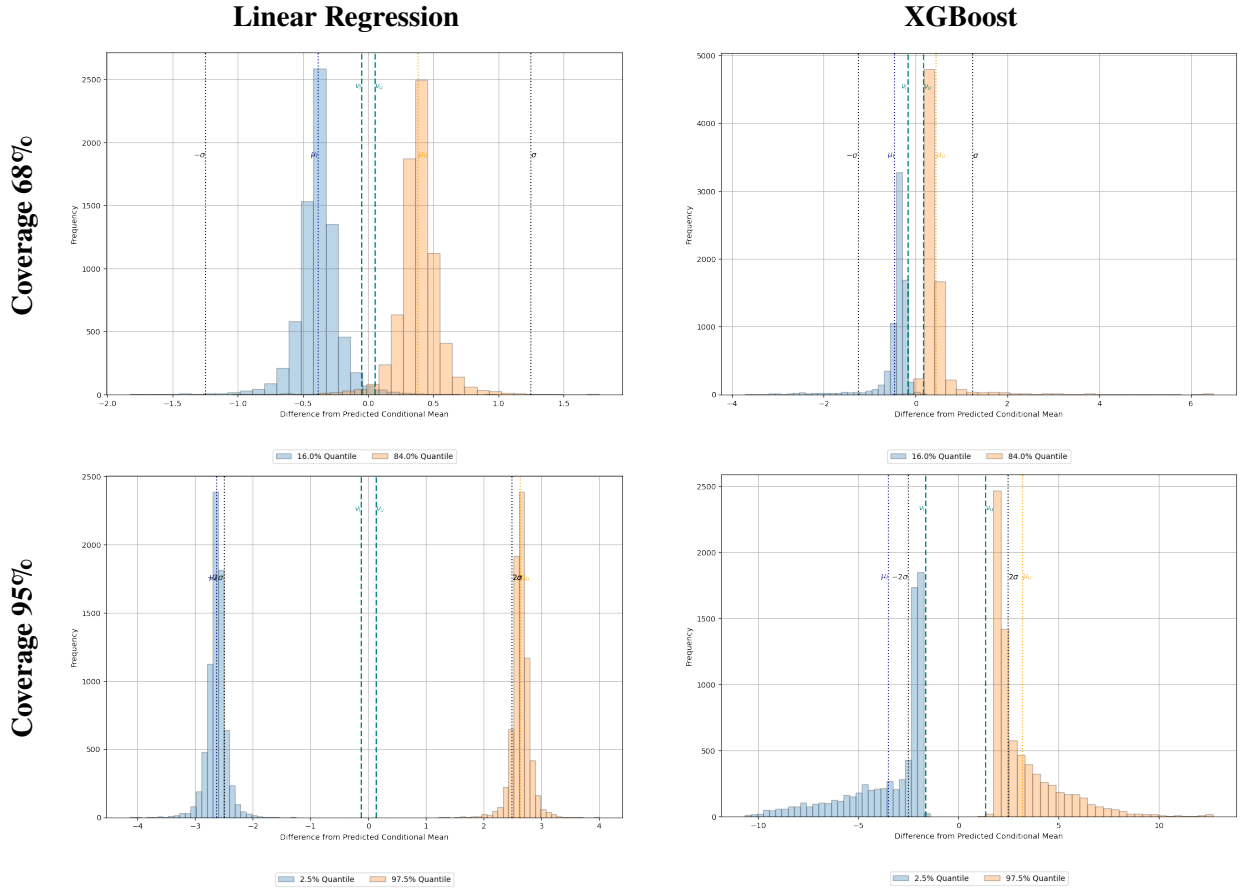


This figure displays the prediction uncertainty for Scope 1 Emissions. The settings are for the learners linear regression and XGBoost and the target coverage rate of 68% and 95%. For better readability, the actual vs. predicted plots display only 2% of the total observations, randomly selected.

Figure 9. Prediction Uncertainty in Different Settings for Scope 1 Emissions

spread interval lengths.

This figure also presents alternative uncertainty measures for comparison. In addition to the conformalized quantile-based measure, a more naive approach is shown using simple one- and two-standard deviations (σ) from the conditional mean, which are typically used for coverage levels 68% and 95%. This naive approach results in significantly larger intervals compared to the conformalized quantile-based method in most settings. In addition, a third method is illustrated that uses (standard) quantile regression (ν). This method underestimates the uncertainty compared to our conformalized approach most of the time. This leads us to conclude that the conformalized quantile-based method is well suited to test prediction uncertainty.



This figure displays the deviation of actual values from the predicted conditional mean for Scope 1 Emissions in different settings. The settings are for uncertainty estimates based on linear regression and XGBoost and for targeted coverage rates of 68% and 95%. In all settings, the conditional mean is predicted by the best-performing XGB meta model. The black dotted lines represent the prediction intervals using one and two standard deviations (σ), respectively. The green dashed lines represent the mean lower (upper) quantile prediction from standard quantile regression, ν_l (ν_u). Finally, the blue and orange bars show the distribution of quantile predictions for the lower and upper quantile from conformalized quantile regression. The mean of lower (upper) predictions is represented by the blue and orange dotted lines, μ_l (μ_u).

Figure 10. Deviation from Conditional Mean for Scope 1 Emissions

The results of the uncertainty measures for other target variables than Scope 1 emissions show consistent results for environmental variables; however, for the variable representing the female board share (see Appendix D), which is truncated between zero and one, the uncertainty distributions differ, reflecting the specific characteristics of the data.

5. Inference with scarce sustainability data and abundant financials

In corporate and sustainability research, empirical analyzes often aim to identify and quantify systematic associations between firm characteristics and sustainability outcomes. To illustrate how our framework can

be applied in empirical research, we consider a stylized example, asking whether younger firms are systematically "greener" than older ones, as reflected in their direct greenhouse gas emissions. While simplified, this setting exemplifies the type of empirical inquiry that arises when analyzing firm-level sustainability outcomes based on observable firm characteristics. Formally, we consider the following linear equation as an illustrative specification:

$$\Delta DirectEmissions_i = \beta_0 + \beta_1 FirmAge_i + Z_i' \beta_Z + \varepsilon_i, \quad (1)$$

where $\Delta DirectEmissions_i$ denotes the deviation of emissions from the sectoral mean of firm i , $FirmAge_i$ is the firm i 's age, and Z_i is a vector of additional covariates of firm i . The coefficient β_1 is the parameter of interest, capturing the average conditional association between firm age and emissions, and β_Z the corresponding coefficient vector for the additional covariates. The term ε_i captures unobserved determinants of emissions.⁶

Empirical research on such associations typically operates in a 'sparse-labels, rich-features' environment: high-quality sustainability outcomes (here, reported emissions) exist only for a subset of companies, while potential covariates—financial and fundamental variables from balance sheet, profit-and-loss statement, and industry classification—are widely available across companies. Estimating associations only on the labeled subset and with a limited set of theory-driven covariates leaves substantial unexplained outcome variation. Consequently, the sampling variance of the estimator $\hat{\beta}_1$ inflates, yielding wide confidence intervals and imprecise estimates of the parameter of interest.

Recent methodological advances in semiparametric and, in particular, prediction-powered inference provide a principled response to this problem (Angelopoulos et al., 2023, 2024; Zrnic & Candès, 2024). In the original prediction-powered inference framework, model predictions are obtained out of sample from a separately trained model, ensuring independence between prediction values and the labeled residuals used for inference. The key mechanism is that if the prediction model explains a substantial fraction of the outcome variance, its out-of-sample predictions can be combined with residuals from labeled data to form unbiased estimating

⁶In our illustrative example, the outcome is the natural logarithm of $(1 + DirectEmissions_i)$ demeaned by the corresponding NACE-2 sector mean. We use a low-dimensional covariate vector, $Z_i' = (TotalAssets_i, TotalDebt_i)$, which yields a (super)population coefficient of determination of $R^2 = 0.092$. We do not impose knowledge of the true functional form between company age and emissions. Accordingly, the parameter of interest β_1 is interpreted as the best linear approximation to the conditional mean of emissions given company age and the selected covariates in the (super)population.

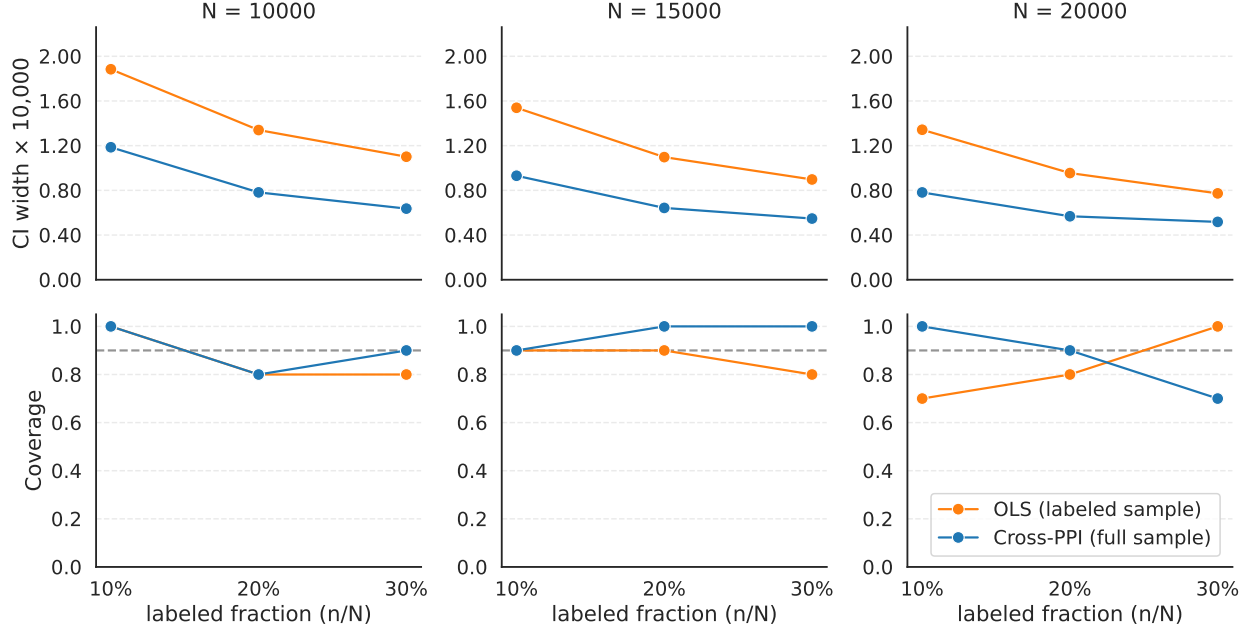
equations with smaller variance. When the prediction model is accurate, the confidence intervals for the parameter of interest (β_1) become narrower. When it is uninformative, the procedure asymptotically reverts to classical OLS inference. Cross-fitted variants of prediction-powered inference (Cross-PPI) further enhance data efficiency by replacing the separate training set with K -fold out-of-fold predictions which preserve the required independence condition while utilizing all available data (Zrnic & Candès, 2024).

We apply the prediction-powered framework to assess whether efficient inference can be achieved for firm-level sustainability relationships when predictive models are trained exclusively on financial and fundamental data. The experiment is implemented in the illustrative setting of equation (1). We use the 2022 cross-section of 20,053 firm-year observations underlying our emission-estimation framework (Section 2), comprising a subset of 75 financial and fundamental features and corresponding direct (Scope 1) emissions.⁷ To benchmark coverage and interval width, we adopt a superpopulation approach. We draw a large bootstrap sample ($N = 100,000$) from the cross-section to define the population parameter β_1^* via OLS on that sample. All reported coverage metrics refer to confidence intervals computed on separate evaluation samples relative to this fixed benchmark β_1^* .

The simulation scenarios vary the total number of observations available $N \in 10,000, 15,000, 20,000$, and set the labeled share at $s \in 0.10, 0.20, 0.30$. For each configuration, we draw a bootstrap sample of size N , treat $n = sN$ companies as labeled, and consider the remaining $N - n$ observations as unlabeled. The classical estimator fits OLS on labeled data and constructs confidence intervals. The estimator based on cross-prediction follows (Zrnic & Candès, 2024) and uses five-fold cross-fitting to generate out-of-fold predictions for labeled observations and fold-averaged predictions for unlabeled ones. The prediction model is the best-performing base learner in our framework—gradient-boosted trees (XGB) minimizing mean squared error—trained only on financial and fundamental covariates. All steps, except for the final regressor, are fitted jointly on labeled and unlabeled samples within each replication to preserve a consistent feature representation. Each configuration is replicated 10 times to approximate empirical coverage.

Figure 11 summarizes the results. The upper panels report the mean width of the confidence intervals for the age coefficient (scaled by 10,000), while the lower panels show empirical coverage of nominal $(1 - \alpha)$ intervals with a target of 0.90 ($\alpha = 0.10$). Across all sample sizes and label shares, Cross-PPI delivers consistently narrower confidence intervals than classical OLS. Empirical coverage is broadly on target for both

⁷We use a subset of financial and fundamental features based on availability for computational efficiency.



This figure shows the confidence intervals for classical OLS (labelled sample) and Cross-PPI (full sample). In the top row, we present mean confidence-interval width (scaled by 10,000). In the bottom row, the empirical coverage with target line at 0.9 are presented. Columns vary $N \in 10,000, 15,000, 20,000$; the horizontal axis shows labelled fractions $n/N \in 0.10, 0.20, 0.30$. The target β_1^* is defined on an independent superpopulation resample. Intervals are computed on separate evaluation samples.

Figure 11. Prediction Powered Inference vs Classical OLS

methods across configurations; the minor deviations from the nominal level are consistent with finite-sample fluctuations. Overall, the results demonstrate that our machine-learning framework—trained exclusively on financial and fundamental data—combined with prediction-powered inference can substantially increase statistical efficiency when sustainability labels are scarce and financial covariates are abundant, the empirically most relevant regime for firm-level sustainability research.

6. Conclusion

This study demonstrates that ML can effectively generate firm-level sustainability data using readily available financial and fundamental metrics only, but generating a variety of metrics, significantly expanding opportunities in sustainability-related decision making and research. Our framework particularly addresses three critical research challenges:

First, our method extends the geographical and temporal scope for researchers. Currently, sustainability datasets are temporally constrained due to relatively recent expansions in reporting standards (Krueger et al., 2024) and in terms of the diversity of sustainability indicators. Consequently, longitudinal or global analy-

ses, such as evaluating the impact of international climate agreements such as the 2015 Paris Agreement, are often hampered by insufficient historical data and self-selection through voluntary reporting. Our ML-driven methodology enables researchers to generate global historical estimates, thereby facilitating more comprehensive and reliable studies. By broadening the range of robustly measurable sustainability metrics, our method enables researchers to conduct nuanced analyses across previously underrepresented sustainability dimensions such as biodiversity or water stress without relying on generalized or opaque third-party estimates. This would help answer questions such as: How effective are climate policies and commitments by companies and investors? How do markets price biodiversity risks?

Second, the introduction of prediction uncertainty measures allows users of sustainability data to apply our framework with greater confidence. Although most emission estimation techniques still rely on point estimation, our framework can guide decision-makers in deriving confidence around their analysis based on our framework. This introduces a new dimension to decision making using estimated sustainability data.

Third, we demonstrate how our ML framework can be combined with recent advances in semiparametric inference to enable valid and efficient estimations of firm-level sustainability relationships when sustainability data are sparse. Our results show that this combination substantially tightens confidence intervals and improves the identification of systematic associations relative to classical estimators relying solely on observed data.

Beyond these methodological contributions, our findings carry significant practical implications for policymakers, companies, and financial institutions. Policymakers may reconsider reliance exclusively on directly reported sustainability data, recognizing that ML-generated data can offer an economically feasible and high-quality alternative when direct data collection proves too costly or impractical. However, regulatory frameworks need to evolve to address the inherent uncertainties and potential biases in predictive modeling, which require higher transparency standards similar to those emerging for ESG ratings in the European Union (General Secretariat of the Council, 2024).

In addition, our analysis underscores the importance of adapting ML models to specific contexts, recognizing variations in performance between sectors, space, and time. Clear documentation of model adaptations, including explicit disclosures about loss-function modifications and training datasets, is essential to maintain transparency and enable users to interpret and trust ML-generated data.

It is important to acknowledge the limitations of our research, particularly regarding generalizability. Our models, developed primarily from data provided by larger and publicly reporting companies, may exhibit reduced accuracy when applied to small and medium-sized enterprises (SMEs), non-profits, or governmental entities.⁸ Additionally, voluntary ESG reporting practices can introduce sampling biases that must be acknowledged in interpreting predictions. Future research should explore methodologies to adapt ML frameworks to these less represented contexts and examine approaches to mitigate biases resulting from non-mandatory reporting practices.

In conclusion, this research demonstrates the considerable potential of machine learning to improve the accessibility, scope, and reliability of sustainability data, supporting informed decision making by companies and policy makers, and also allowing researchers to draw more reliable and meaningful conclusions.

⁸In Appendix B, we show that the prediction performance of our models for smaller companies tends to be worse.

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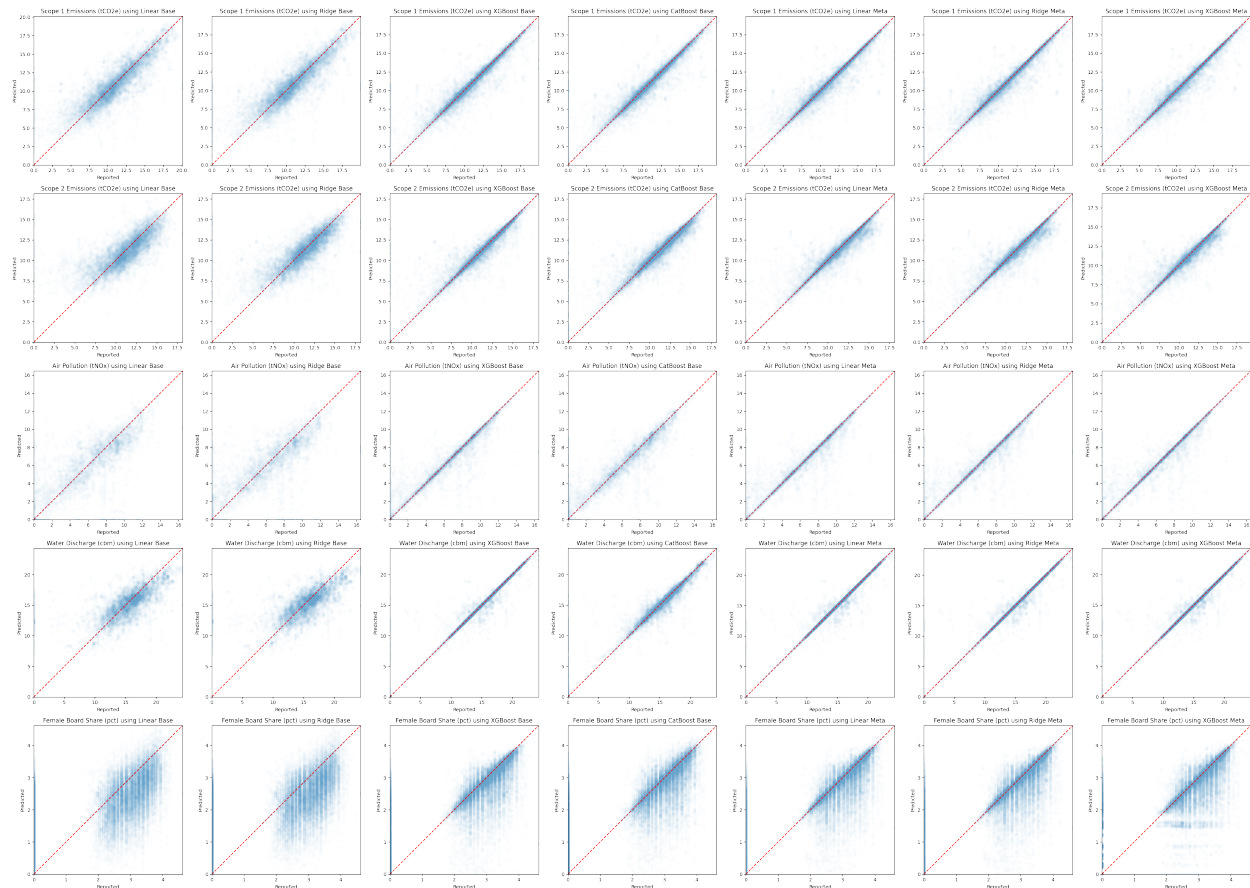
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A. Model performance: reported vs. predicted values

Figure 12 presents a comparison between reported and predicted values within our test data set for both baseline and meta models in the five sustainability metrics. This comparison reaffirms that more complex models, such as XGBoost and CatBoost, as well as the meta models, consistently outperform linear models like OLS and Ridge by a significant margin.



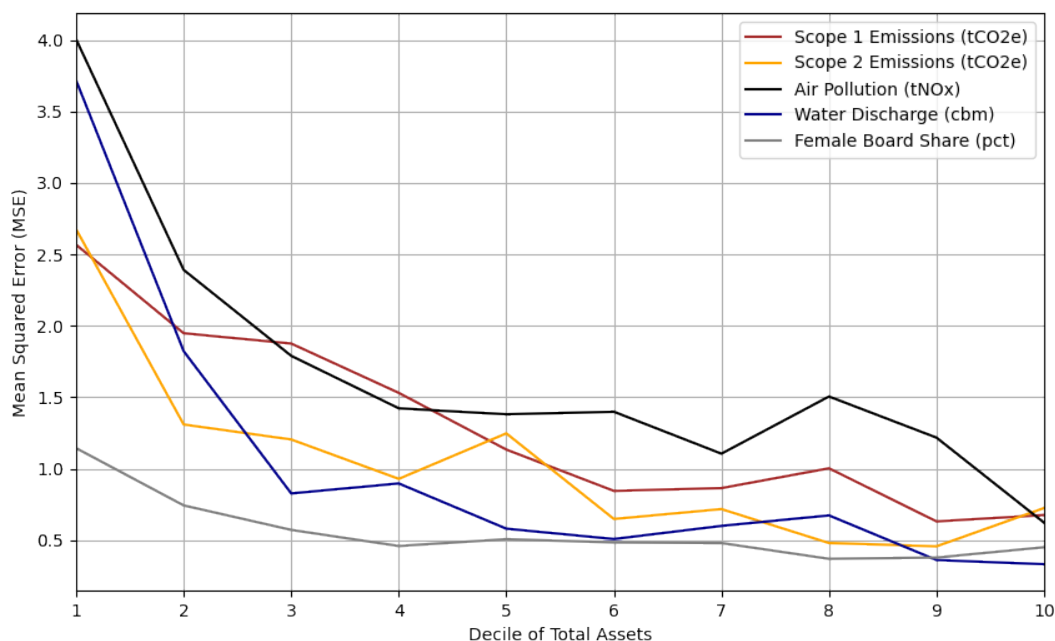
This figure displays the plots of the reported vs predicted variables for the five target variables vertically (Scope 1 emissions, Scope 2 emissions, air pollution, water discharge, and female board share) and the best performing model per learner horizontally (linear regression, ridge, CATBoost, XGBoost, meta-linear, meta-ridge, meta-XGBoost).

Figure 12. Plots reported vs predicted for best model per target variable and learner

B. Model performance by company size

Figure 13 illustrates the variation in model performance relative to the size of the company, represented by deciles of the size of the company and the corresponding MSE per decile. The results indicate that model performance is notably poorer in the lowest decile, which includes the smallest companies. Despite this, the

performance remains superior compared to previous studies even in this decile. Following the initial decile, there is a sharp improvement in model performance, which then levels off with only minor variations around deciles 7 to 9.

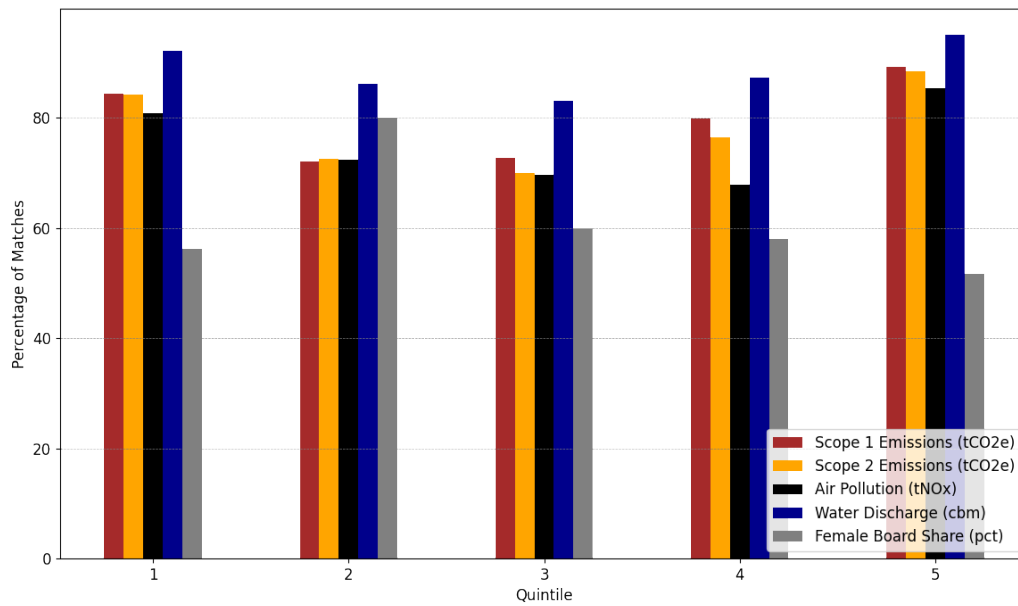


This figure displays the MSE of model predictions for company-size deciles for the five target variables: Scope 1 emissions, Scope 2 emissions, air pollution, water discharge, and female board share.

Figure 13. Model Performance by Company Size

C. Quintile analysis of sustainability metrics

Figure 14 presents the quintile results, also discussed in the main body of the paper, showing the percentage of matches for each sustainability metric within their respective quintile brackets. The primary finding is that the variation in model fit across quintiles is generally minimal for most sustainability metrics, with the exception of the female board share metric, where substantial variation is observed among the quintiles. This suggests that the predictive accuracy for the female board share is more sensitive to the distribution of data between different quintiles compared to other sustainability metrics.

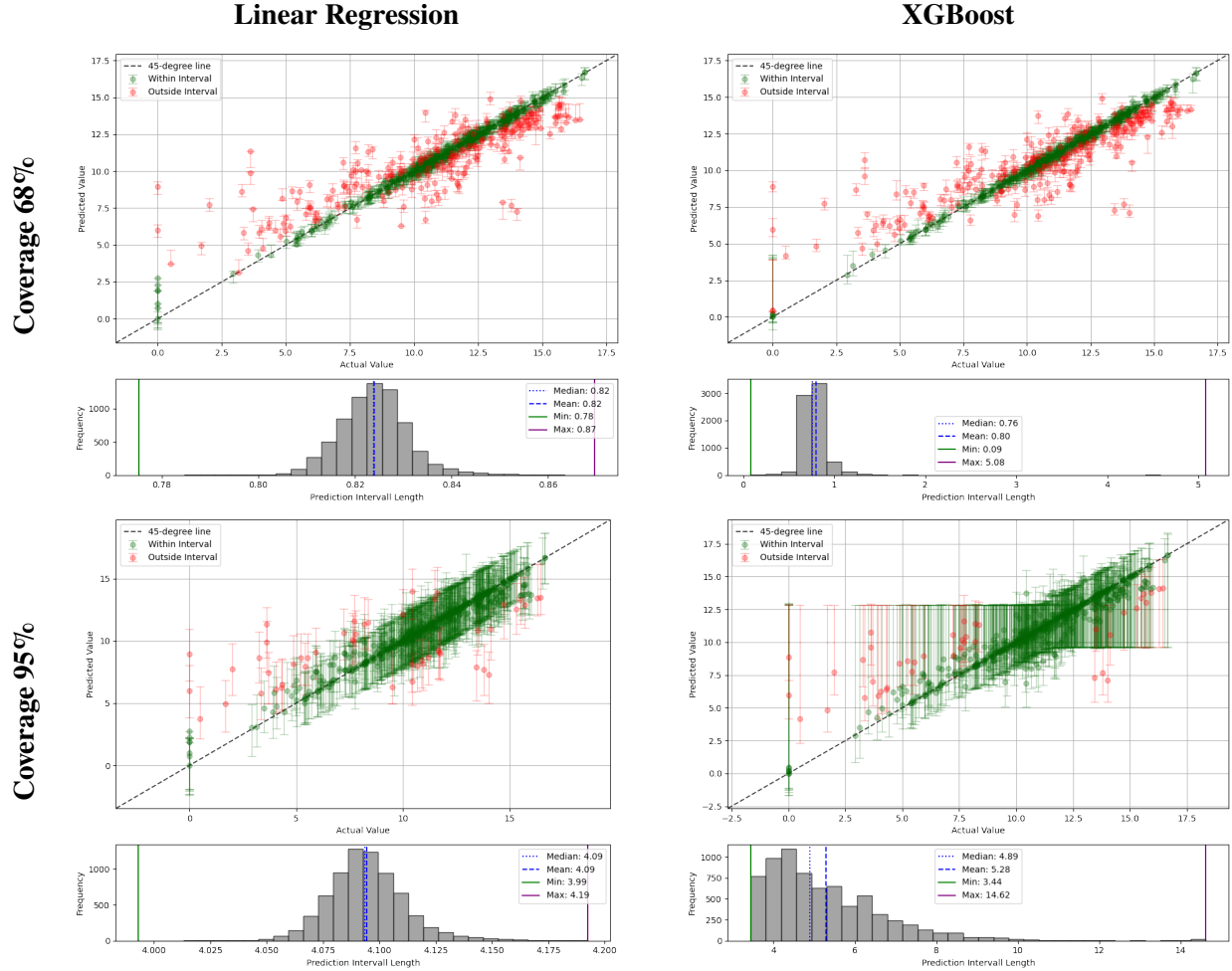


This figure displays the percentage of correctly allocated observations (predicted vs actual) per quintile for the five target variables: Scope 1 emissions, Scope 2 emissions, air pollution, water discharge, and female board share.

Figure 14. Percentage of Quintile Matches

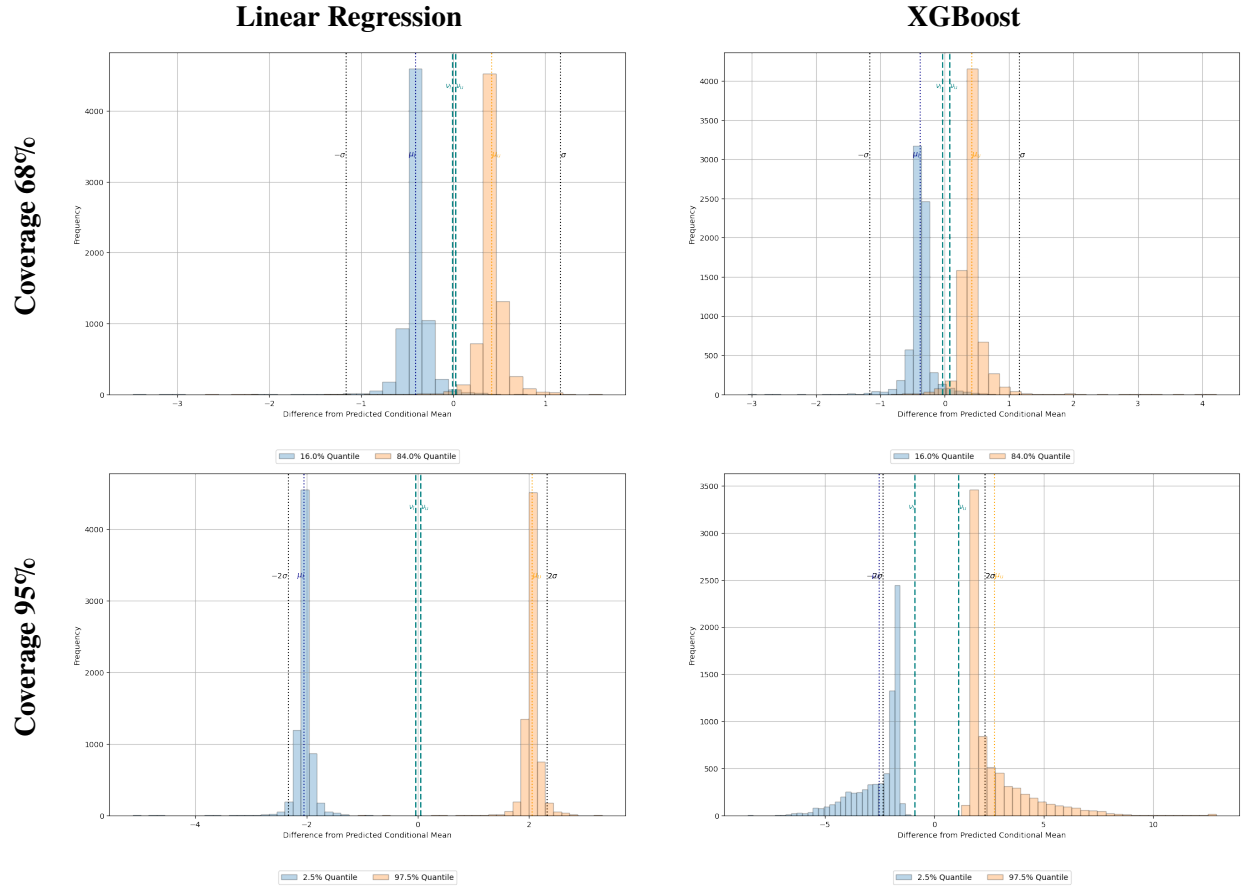
D. Prediction uncertainty for remaining target variables

Scope 2 Emissions



This figure displays conformalized prediction uncertainty for Scope 2 Emissions. The settings are for the learners linear regression and XGBoost and the coverage of 68% and 95%. For better readability, the figure shows 2% of total observations which are randomly selected.

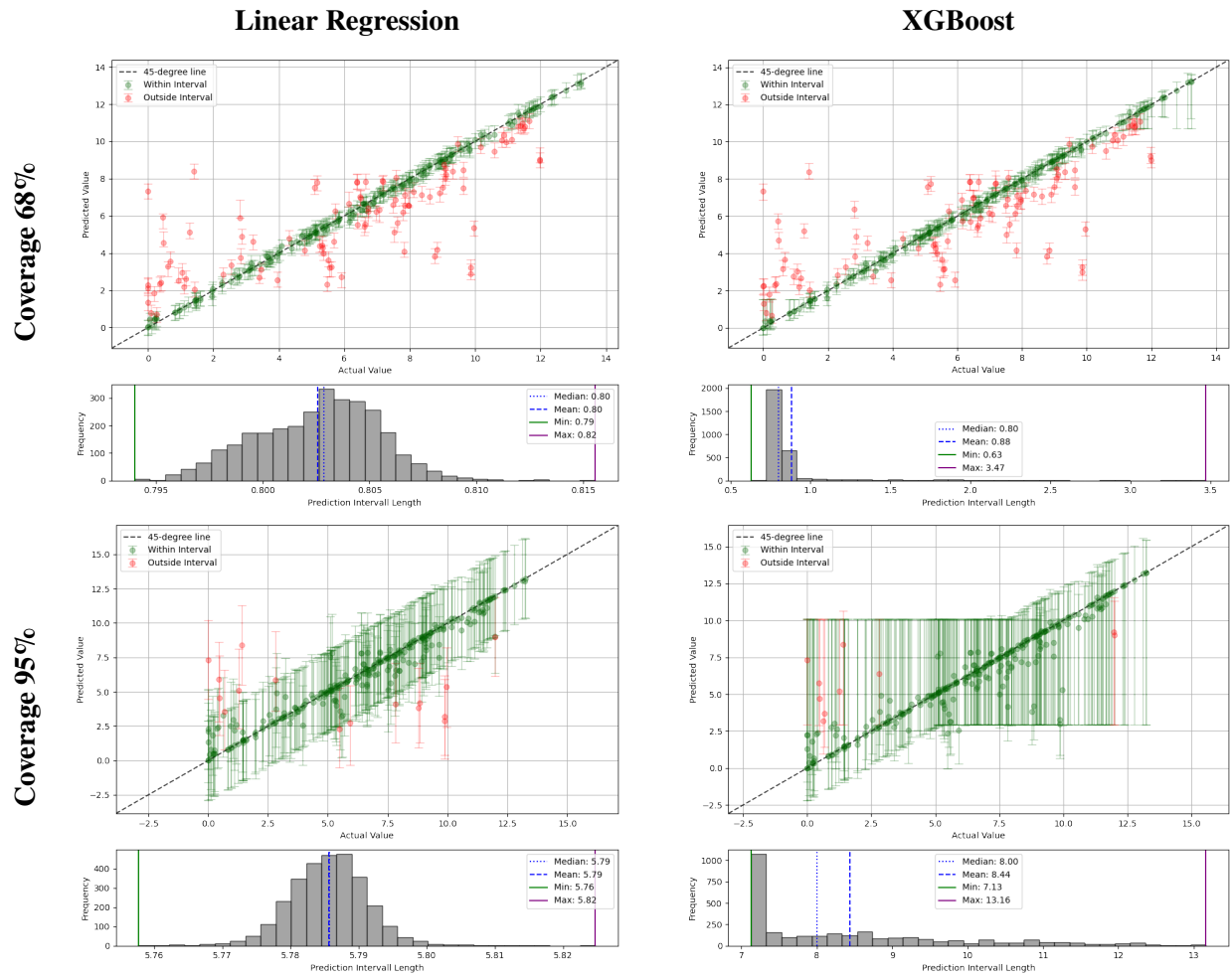
Figure 15. Conformalized Prediction Uncertainty in Different Settings for Scope 2 Emissions



This figure displays the deviation for the conditional mean for Scope 2 Emissions in different settings. The settings are for uncertainty estimates based on linear regression and XGBoost and for targeted coverage rates of 68% and 95%. In all settings, the conditional mean is predicted by the best-performing XGB meta model. The black dotted lines represent the prediction intervals using one and two standard deviations (σ), respectively. The green dashed lines represent the mean lower (upper) quantile prediction from standard quantile regression, v_l (v_u). Finally, the blue and orange bars show the distribution of quantile predictions for the lower and upper quantile from conformalized quantile regression. The mean of lower (upper) predictions is represented by the blue and orange dotted lines, μ_l (μ_u).

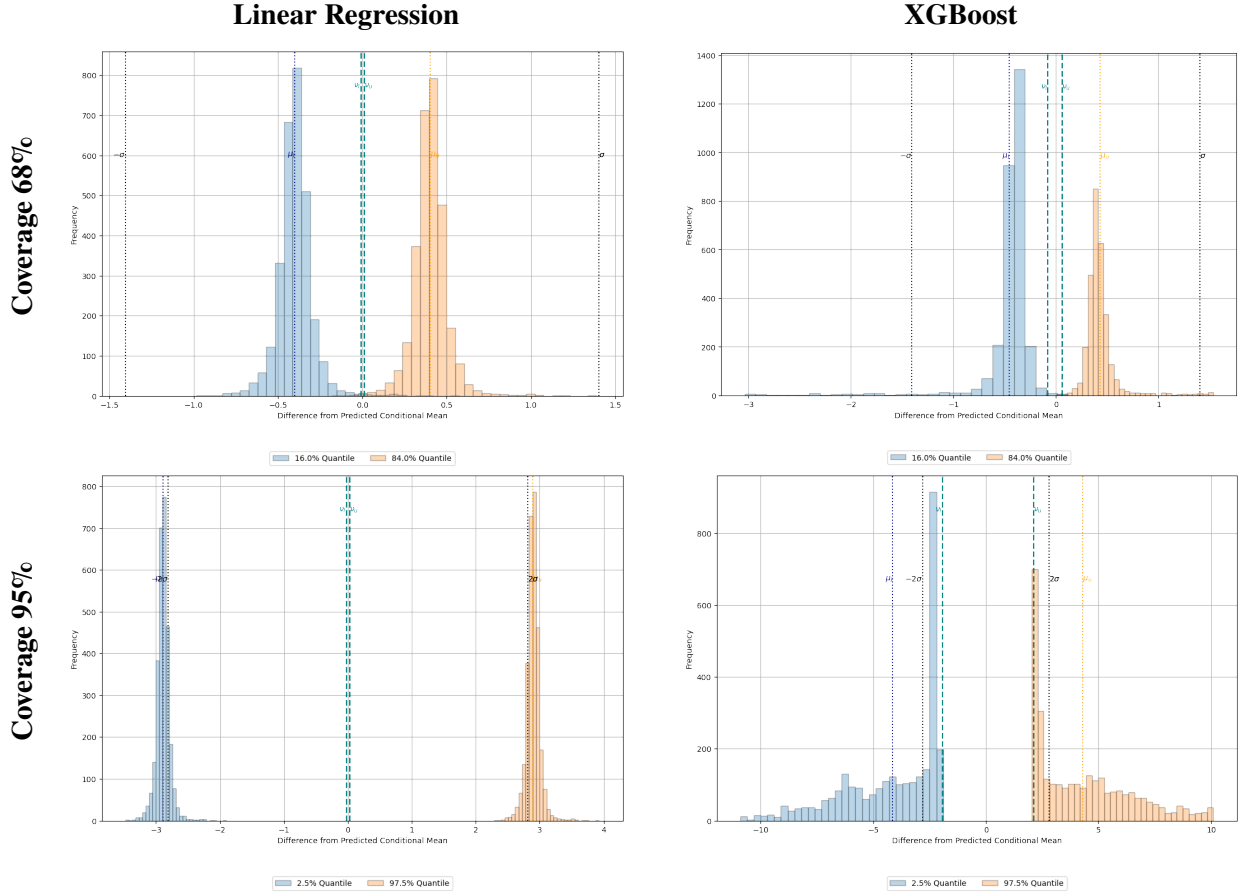
Figure 16. Deviation from Conditional Mean for Scope 2 Emissions

Air Pollution



This figure displays conformalized prediction uncertainty for NOx Emissions. The settings are for the learners linear regression and XGBoost and the coverage of 68% and 95%. For better readability, the figure shows 2% of total observations which are randomly selected.

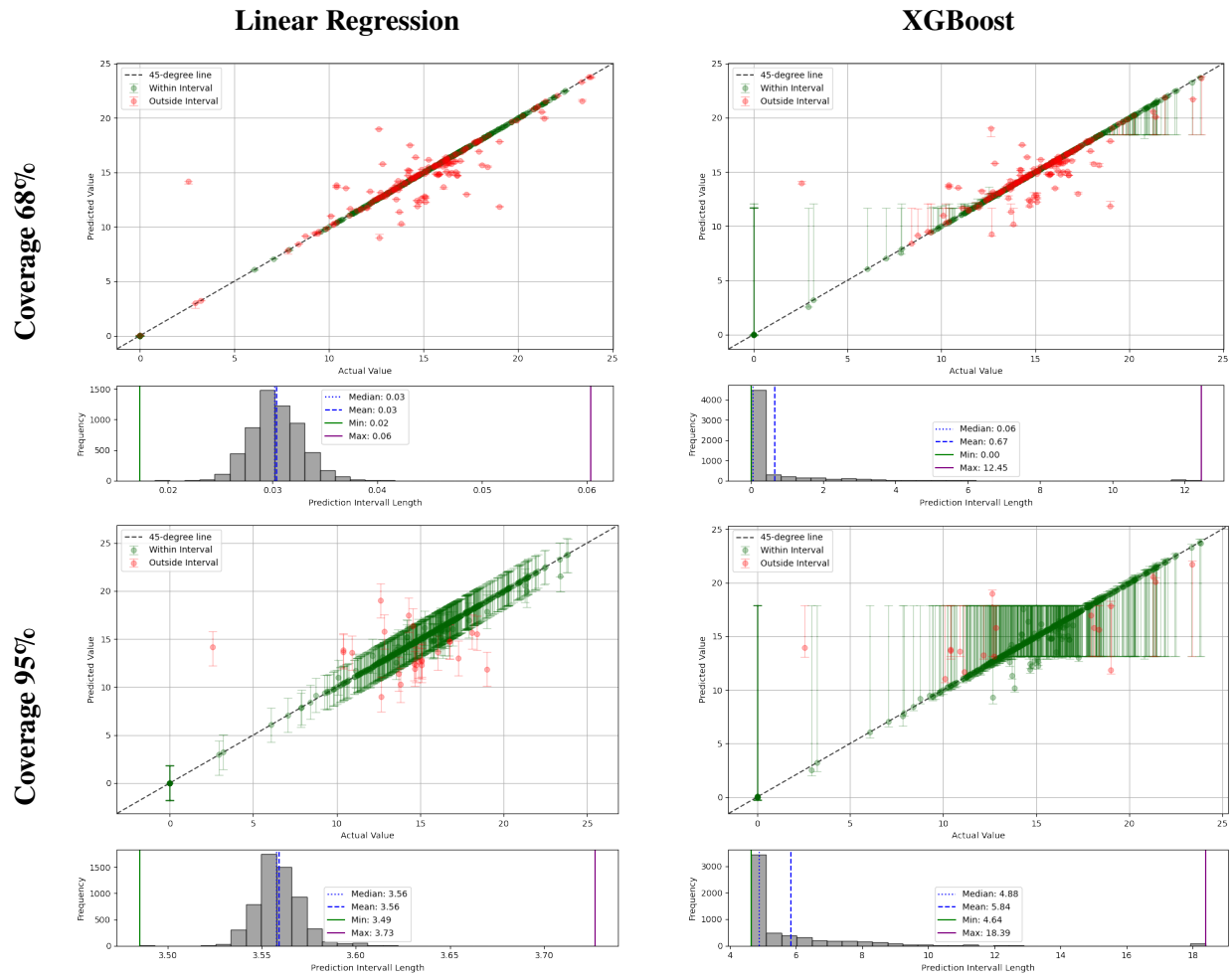
Figure 17. Conformalized Prediction Uncertainty in Different Settings for NOx Emissions



This figure displays the deviation for the conditional mean for NOx Emissions in different settings. The settings are for uncertainty estimates based on linear regression and XGBoost and for targeted coverage rates of 68% and 95%. In all settings, the conditional mean is predicted by the best-performing XGB meta model. The black dotted lines represent the prediction intervals using one and two standard deviations (σ), respectively. The green dashed lines represent the mean lower (upper) quantile prediction from standard quantile regression, v_l (v_u). Finally, the blue and orange bars show the distribution of quantile predictions for the lower and upper quantile from conformalized quantile regression. The mean of lower (upper) predictions is represented by the blue and orange dotted lines, μ_l (μ_u).

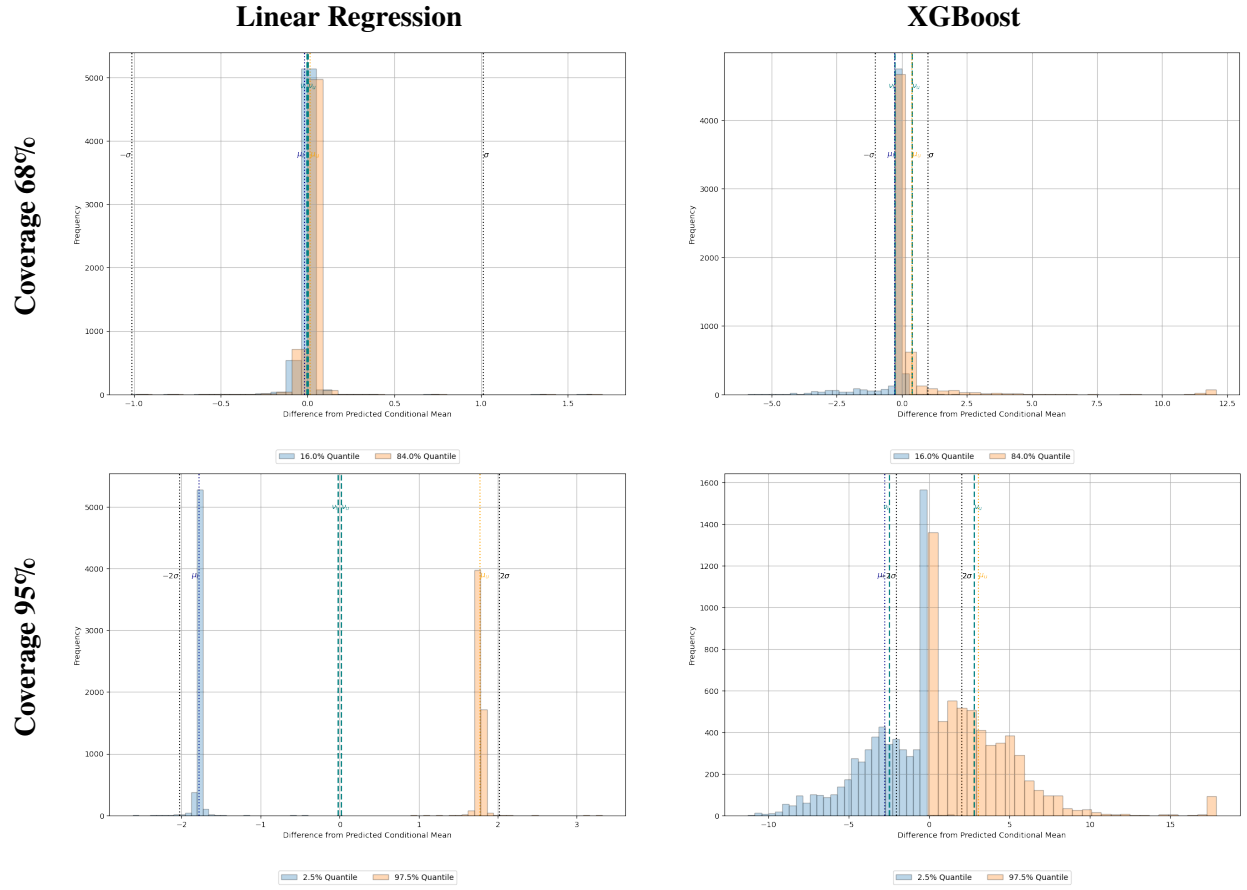
Figure 18. Deviation from Conditional Mean for NOx Emissions

Water Discharge



This figure displays conformalized prediction uncertainty for Water Discharge. The settings are for the learners linear regression and XGBoost and the coverage of 68% and 95%. For better readability, the figure shows 2% of total observations which are randomly selected.

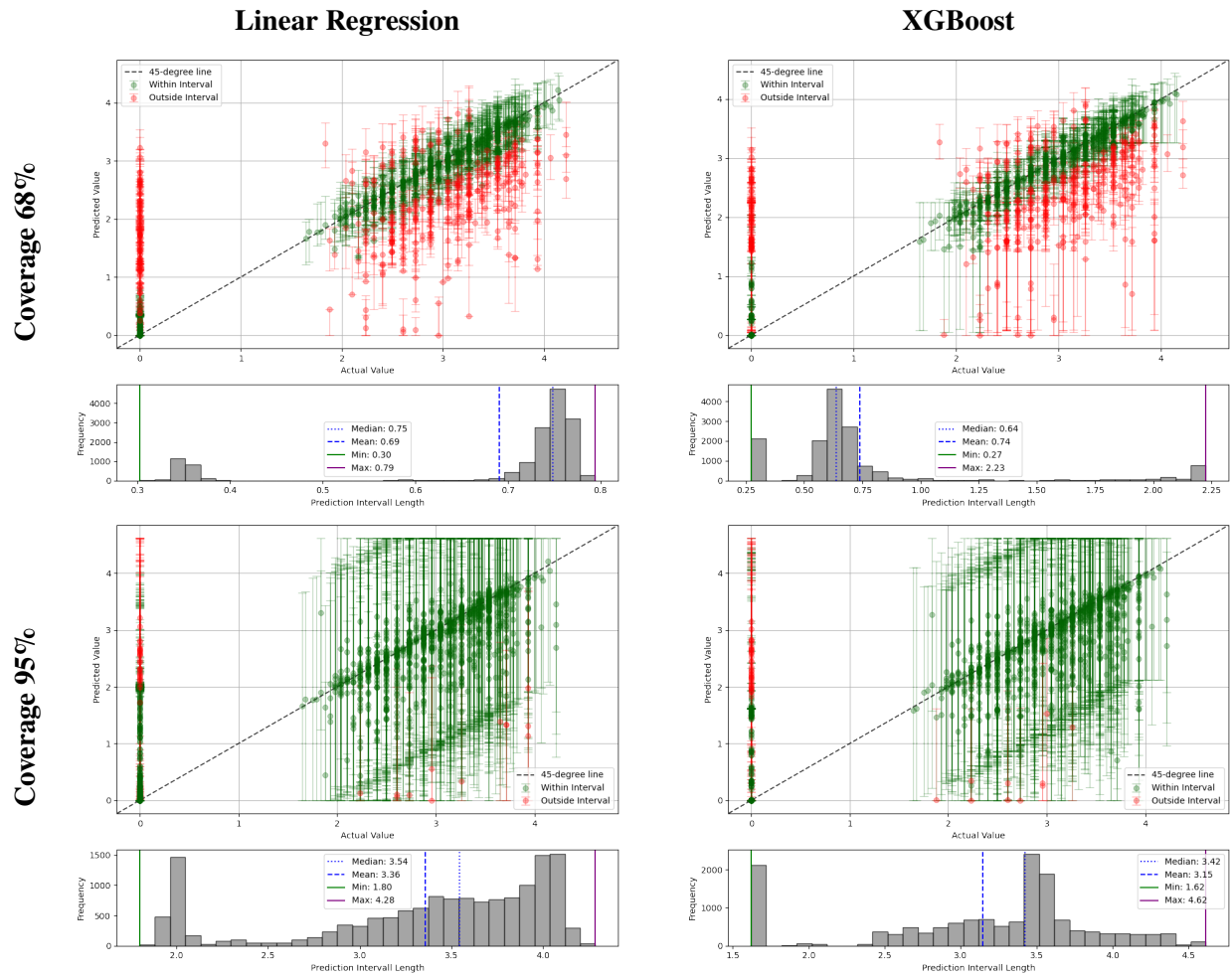
Figure 19. Conformalized Prediction Uncertainty in Different Settings for Water Discharge



This figure displays the deviation for the conditional mean for Water Discharge in different settings. The settings are for uncertainty estimates based on linear regression and XGBoost and for targeted coverage rates of 68% and 95%. In all settings, the conditional mean is predicted by the best-performing XGB meta model. The black dotted lines represent the prediction intervals using one and two standard deviations (σ), respectively. The green dashed lines represent the mean lower (upper) quantile prediction from standard quantile regression, ν_l (ν_u). Finally, the blue and orange bars show the distribution of quantile predictions for the lower and upper quantile from conformalized quantile regression. The mean of lower (upper) predictions is represented by the blue and orange dotted lines, μ_l (μ_u).

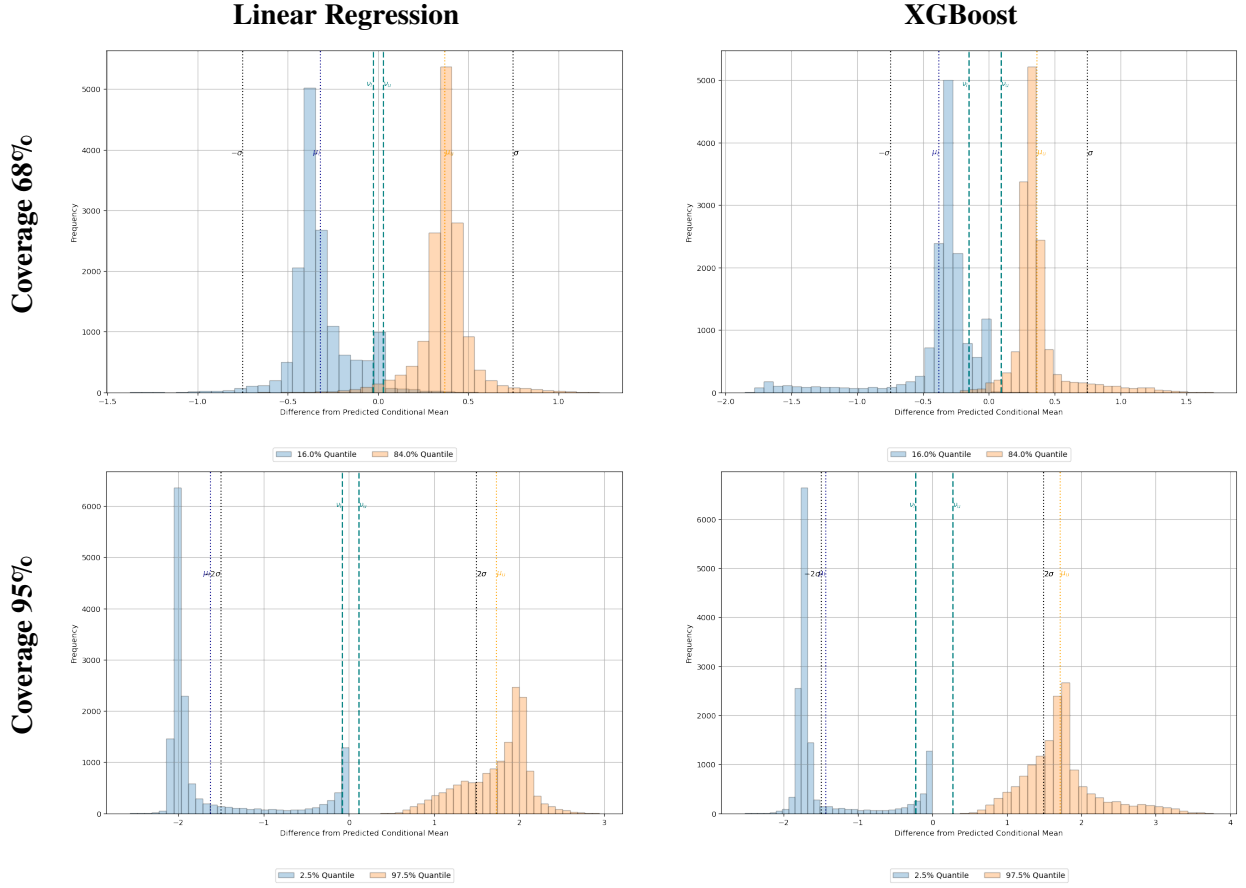
Figure 20. Deviation from Conditional Mean for Water Discharge

Female Board Share



This figure displays conformed prediction uncertainty for Female Board Share. The settings are for the learners linear regression and XGBoost and the coverage of 68% and 95%. For better readability, the figure shows 2% of total observations which are randomly selected.

Figure 21. Conformed Prediction Uncertainty in Different Settings for Female Board Share



This figure displays the deviation for the conditional mean for Female Board Share in different settings. The settings are for uncertainty estimates based on linear regression and XGBoost and for targeted coverage rates of 68% and 95%. In all settings, the conditional mean is predicted by the best-performing XGB meta model. The black dotted lines represent the prediction intervals using one and two standard deviations (σ), respectively. The green dashed lines represent the mean lower (upper) quantile prediction from standard quantile regression, v_l (v_u). Finally, the blue and orange bars show the distribution of quantile predictions for the lower and upper quantile from conformalized quantile regression. The mean of lower (upper) predictions is represented by the blue and orange dotted lines, μ_l (μ_u).

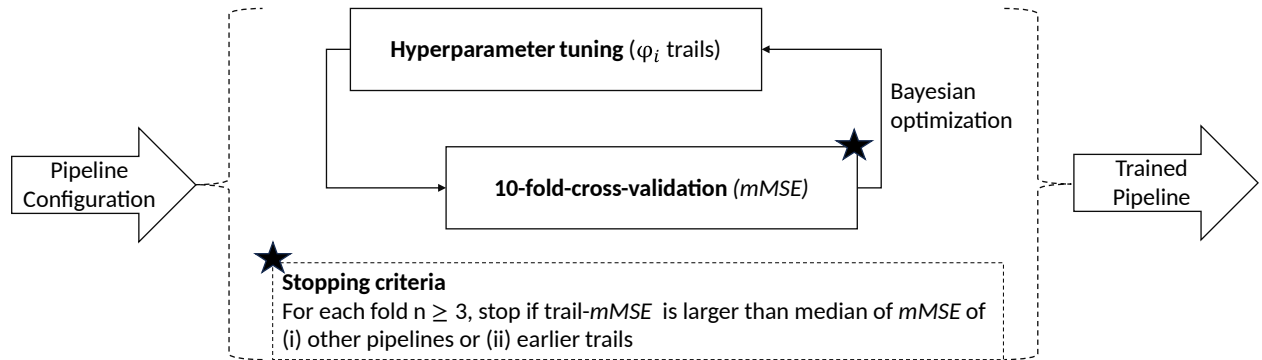
Figure 22. Deviation from Conditional Mean for Female Board Share

E. Early stopping

In our efforts to improve computational efficiency during model training, we implemented an early stopping mechanism. This mechanism is designed to reduce unnecessary computations during the hyperparameter tuning process, specifically within the ten-fold cross-validation phase of each trial. Figure 23 visually represents the early stopping process.

The early stopping framework is structured as follows:

1. **Initialization of Trials:** During the hyperparameter tuning process, each trial undergoes a tenfold cross-validation to evaluate the model's performance under different parameter settings.
2. **Minimum Trial Requirement:** To ensure that the model has sufficient opportunity to learn and stabilize, the early stopping mechanism is only considered after the first three folds of the cross-validation process have been completed.
3. **Performance Comparison:** After the third fold, the mechanism compares the MSE obtained in the current fold with the median MSEs of previously evaluated models or earlier trials of the same model configuration.
4. **Activation of Early Stopping:** If the MSE of the current trial exceeds the average MSEs of the previous trials, the early stopping mechanism is triggered. When activated, this mechanism halts the remaining cross-validation folds for the current trial and proceeds directly to the next model configuration. This avoids further computation on a model configuration that is unlikely to outperform existing configurations.



This figure displays the early stopping mechanism that is applied throughout the code.

Figure 23. Early Stopping Framework

F. Computational efficiency vs. comprehensiveness trade-off in model development

In this section, we elaborate on the trade-offs we encountered between computational efficiency and the comprehensiveness of our framework in model development.

Our framework to model configuration deliberately incorporates a trade-off between computational demands and the comprehensiveness of the analysis. By treating preprocessing steps as additional hyperparameters and employing a wide range of models and configurations, we place a significant burden on computational resources. This strategy, while resource-intensive, is aimed at enhancing the replicability of our results and ensuring that our methods are broadly applicable, independent of specific institutional contexts in which the data are generated.

The versatility offered by this framework can be valuable to the research community, as it allows for more generalized and adaptable modeling techniques.

However, this versatility comes at the cost of computational efficiency. Despite implementing early stopping mechanisms to reduce unnecessary calculations, the overall process remains resource-demanding. In the following, we provide an overview of the computational resources required for our framework:

- CPU: **2 x AMD EPYC Milan 7713 - 64-Core**
- GPU: **NVIDIA RTX Ada A6000**
- Time Consumption: **19.01 days** (*assuming serial execution*)
- Electricity Consumption: **87.94 kWh** (*estimation*)
- Early Stopping Effect: **active in 37.86% of 59,250 trials**

The above specifications highlight the scale of computational resources needed to execute our models effectively. The use of powerful CPUs and GPUs is essential to manage the large datasets and numerous trials involved in our comprehensive framework to discovering best performing models. Even with these resources, the time required to complete the model runs is considerable, emphasizing the trade-off between the depth of analysis and computational efficiency.

The introduction of early stopping mechanisms plays a crucial role in mitigating some of the computational burdens. As detailed in Annex B, early stopping is activated in approximately 37.86% of the 59,250 trials, significantly reducing the time and energy required to train the models. This not only improves efficiency but also ensures that computational resources are allocated more effectively towards promising model configurations.

G. Insights and recommendations for replication or extension of our work

In this annex, we provide guidance for future researchers who aim to replicate or build on our findings with reduced computational effort. By analyzing the performance of various model configurations and the factors that contribute to the best results, we draw several conclusions that can inform more efficient model development in similar studies. In addition, we provide guidance on the use of our data.

Applying our method

Table V summarizes the configurations of the best-performing models.

Target Variable	Missing Indicator		Imputation			Outlier Removal		Scaler		Transformation		Feature Selection	
	No	Yes	Mean	Median	Iterative	None	Winzorize	Standard	Robust	None	Quantile	None	Lasso
Scope 1 Emissions (tCO2e)	6	6	4	6	2	11	1	6	6	5	7	12	0
Scope 2 Emissions (tCO2e)	2	10	5	7	0	5	7	4	8	5	7	12	0
Air Pollution (tNOx)	4	7	5	6	0	8	3	1	10	8	3	11	0
Water Discharge (cbm)	3	9	8	2	2	10	2	4	8	7	5	12	0
Female Board Share (pct)	4	8	5	6	1	4	8	3	9	10	2	12	0

This table summarizes the best performing model setups along the dimensions missing indicator, imputation, outlier removal, scaler, transformation and feature selection for the five target variables: Scope 1 emissions, Scope 2 emissions, air pollution, water discharge, and female board share.

Table V. Summary of Winning Model Configurations

The analysis of the best performing models reveals several insights that can guide future research.

- **Meta Learners and Complex Models:** As stated in the main body of the paper, more complex models, such as meta learners, tend to outperform simpler linear models. Researchers should consider including these advanced techniques in their pipelines to achieve better performance.
- **Pipeline Options:** The study shows that there is no universal best configuration across all preprocessing steps. This suggests that researchers working with different data sets or in different use cases may need to experiment with the full range of pipeline options, rather than relying on a predetermined set of configurations. For example, there is no clear trend that favors a particular missing indicator or scaling

method in all target variables.

- **Feature Selection:** The analysis indicates that in our case, model configurations without feature selection generally performed better. This finding suggests that including more features, rather than reducing them, can enhance model performance. Future researchers should be cautious about overly aggressive feature selection, particularly when working with rich datasets, as it might lead to the loss of valuable information.

In addition to these findings on pipeline building, researchers might want to implement early stopping mechanisms, as discussed in Annex E, to reduce computational burden while ensuring that only the most promising configurations are fully explored.

Using our data

When using our data, researchers should consider model variation and prediction uncertainty.

Understand and adapt to model variations:

- **Temporal Dynamics:** Recognize that the quality of the model predictions may vary over time. For example, earlier periods in datasets may yield better predictions than more recent ones, potentially due to differences in data completeness or quality. Researchers should explore temporal segmentation in their analyses and adjust the models accordingly.
- **Regional Variations:** Consider the regional context of the data. Prediction accuracy may differ significantly across geographic regions, with some regions (e.g., Europe) performing better. Adapt the models to account for regional data availability and variability.
- **Sectoral Differences:** Different sustainability indicators (e.g., water discharge, air pollution) exhibit sector-specific prediction performance. Researchers should analyze sectoral characteristics and adapt models to suit the particularities of each sector.

Researchers may achieve this either by retraining the model (i.e. adapting the loss function to their needs) or by making sure that the model specifications in their research capture those variations.

Use of prediction uncertainty in research:

- **Reliability:** Prediction uncertainty, specifically through conformal prediction intervals, enhances the

reliability of sustainability data by quantifying the confidence level of each prediction. For example, a 95% conditional coverage interval indicates that for a given company, there is a 95% probability that its actual datum lies within the estimated range. This provides researchers with a transparent measure of the quality and trustworthiness of the imputed data, especially in contexts where decision-making depends on accurate sustainability metrics.

- **Robustness:** By explicitly incorporating uncertainty measures, researchers can assess the robustness of their models in various scenarios. For example, when conducting sensitivity analyses or stress tests, prediction intervals provide a range of plausible outcomes rather than relying on single-point estimates, which might be overly optimistic or pessimistic. This allows researchers to account for data imperfections and ensure that their findings hold under different levels of risk tolerance (e.g., using narrower 68% intervals for moderate scenarios and wider 95% intervals for extreme cases).

H. Conformalized quantile prediction and property of the quantile regression loss function

We implement the conformalized quantile prediction in four steps: To implement this approach, we consider three disjoint datasets: Z_{train} , Z_{cal} , and Z_{test} , where each set consists of i.i.d. samples, clustered by company. Let $\mathcal{F}$ denote the set of all companies, and let $Z_F = \{(y_i, x_i) : i \in I_F\}$ be the set of all observations associated with company $F \in \mathcal{F}$, where I_F denoted the index set for company F . The data is partitioned such that: $Z_{train} = \bigcup_{F \in \mathcal{F}_{train}} Z_F$, $Z_{cal} = \bigcup_{F \in \mathcal{F}_{cal}} Z_F$, $Z_{test} = \bigcup_{F \in \mathcal{F}_{test}} Z_F$, where $\mathcal{F}_{train}$, $\mathcal{F}_{cal}$, and $\mathcal{F}_{test}$ form a partition of the set of companies $\mathcal{F}$. The partitioning is done such that approximately 80% of the data is in Z_{train} , and Z_{cal} , Z_{test} each contain approximately 10% of the data.

1. *Quantile Regression:* We train the best base models for each learner as well as the meta models on the training data set using the loss function

$$L_\alpha(\hat{y}_\alpha, y) = \begin{cases} \alpha(y - \hat{y}_\alpha) & \text{if } y > \hat{y}_\alpha \\ (1 - \alpha)(\hat{y}_\alpha - y) & \text{if } y \leq \hat{y}_\alpha \end{cases}$$

with $\alpha \in \{(1 - \tau)/2, (1 + \tau)/2\}$. Here, y is the actual value of the target variable, $\hat{y}_\alpha = \hat{y}_\alpha(x)$ is the quantile prediction of the target variable as a function of the features $\mathbf{x}$.

2. *Quantile Prediction:* We then use the trained models to predict the conditional quantiles $\hat{y}_\alpha(x)$ for each observation in the calibration data set. x is a vector of the predictor variables for each observation.
3. *Conformal Scores:* Based on these quantile predictions, we determine conformal scores, $c(\mathbf{x}, y) = \max\{\hat{y}_{(1-\tau)/2}(\mathbf{x}) - y, y - \hat{y}_{(1+\tau)/2}(\mathbf{x})\}$ for each observation in the calibration data set.
4. *Rectifying Quantiles:* Defining $\hat{r} = \text{Quantile}\left(\frac{\lceil (n_{cal}+1)\tau \rceil}{n_{cal}}, \{c_1, \dots, c_{n_{cal}}\}\right)$ we can derive conditional prediction intervals

$$I(\mathbf{x}) = [\hat{y}_{\tau/2}(\mathbf{x}) - \hat{r}, \hat{y}_{1-\tau/2}(\mathbf{x}) + \hat{r}]$$

for each observation.

We minimize the expected loss function with respect to $\hat{y}_\alpha$ for the quantile level α .

$$\mathbb{E}[L_\alpha(y, \hat{y}_\alpha)] = \alpha \int_{\hat{y}_\alpha}^{\infty} (y - \hat{y}_\alpha) f(y) dy + (1 - \alpha) \int_{-\infty}^{\hat{y}_\alpha} (\hat{y}_\alpha - y) f(y) dy$$

Taking the derivative with respect to $\hat{y}_\alpha$:

$$\frac{\partial \mathbb{E}(\cdot)}{\partial \hat{y}_\alpha} = \alpha \int_{\hat{y}_\alpha}^{\infty} \frac{\partial}{\partial \hat{y}_\alpha} (y - \hat{y}_\alpha) f(y) dy + (1 - \alpha) \int_{-\infty}^{\hat{y}_\alpha} \frac{\partial}{\partial \hat{y}_\alpha} (\hat{y}_\alpha - y) f(y) dy$$

Applying the Leibniz rule for differentiation under the integral sign:

$$\begin{aligned} &= \alpha \left[\int_{\hat{y}_\alpha}^{\infty} \frac{\partial}{\partial \hat{y}_\alpha} (y - \hat{y}_\alpha) f(y) dy + (\hat{y}_\alpha - y) f(y) \Big|_{y=\hat{y}_\alpha} \cdot \frac{d\hat{y}_\alpha}{d\hat{y}_\alpha} \right] \\ &\quad + (1 - \alpha) \left[\int_{-\infty}^{\hat{y}_\alpha} \frac{\partial}{\partial \hat{y}_\alpha} (\hat{y}_\alpha - y) f(y) dy + (\hat{y}_\alpha - y) f(y) \Big|_{y=\hat{y}_\alpha} \cdot \frac{d\hat{y}_\alpha}{d\hat{y}_\alpha} \right] \end{aligned}$$

Simplifying further, we obtain:

$$= -\alpha \int_{\hat{y}_\alpha}^{\infty} f(y) dy + (1 - \alpha) \int_{-\infty}^{\hat{y}_\alpha} f(y) dy$$

Let $F(\hat{y}_\alpha)$ denote the cumulative distribution function (CDF) of $f(y)$:

$$\begin{aligned}
\Rightarrow \frac{\partial \mathbb{E}(.)}{\partial \hat{y}_\alpha} &= -\alpha(1 - F(\hat{y}_\alpha)) + (1 - \alpha)F(\hat{y}_\alpha) \\
&= -\alpha + \alpha F(\hat{y}_\alpha) + F(\hat{y}_\alpha) - \alpha F(\hat{y}_\alpha) \\
&= F(\hat{y}_\alpha) - \alpha
\end{aligned}$$

Setting this derivative to zero to find the minimum:

$$\frac{\partial \mathbb{E}(.)}{\partial \hat{y}_\alpha} = 0 \Rightarrow F(\hat{y}_\alpha) - \alpha = 0$$

Thus, at the optimal point:

$$F(\hat{y}_\alpha) = \alpha$$

I. Summary statistics predictors

Table VI. Summary statistics of predictor variables

The table presents summary statistics for the predictor variables. These statistics include the count of observations, the mean, standard deviation, minimum, maximum, and percentiles (25%, 50%, 75%) for each variable. The units for each variable are indicated in the "Unit" column, with common abbreviations such as B for billions, M for millions, and K for thousands.

	count	mean	std	min	25%	50%	75%	max	Unit
Accounts Payable - Long Term	18316	38.90	467.62	-7.88	0.01	0.11	0.93	18479.00	B
Avg. Payables Payment Days	80527	303.08	23949.38	-32528.25	35.47	56.43	93.25	5769517.74	Unit
Brands, Patents, Trademarks, Marketing (Gross)	27924	8.91	112.41	-0.02	0.01	0.08	0.49	5118.16	B
Capital Expenditures (Total)	101362	79.14	1009.41	-4.17	0.05	0.34	2.94	75161.53	B
Cash & Cash Equivalents	94014	155.92	1722.84	-7.43	0.10	0.66	6.18	110763.21	B
Cash & Cash Equivalents (Total)	103303	223.49	2980.01	-7.43	0.11	0.79	8.87	202104.93	B
Cash & Short-Term Investments	98451	222.83	2579.73	-0.01	0.15	0.95	9.98	124652.84	B
Computer Software (Net)	32490	11.22	113.29	-527.61	0.01	0.07	0.59	5744.00	B
Long Term Debt Issued (Cash Flow)	59479	193.77	2415.98	-111.28	0.08	0.85	8.18	145183.14	B
Long & Short Term Debt Issued (Cash Flow)	19552	1839.72	48455.29	-24.76	0.10	0.95	5.65	1787877.77	B
Short Term Debt Issued (Cash Flow)	11008	471.55	4511.91	-12.11	0.02	0.51	17.66	192778.56	B
Long Term Debt (Total)	96939	342.28	3355.23	-1302.34	0.23	1.95	15.46	183775.07	B
Total Debt	99726	556.41	4952.84	-0.12	0.35	2.78	24.86	205362.30	B
Depreciation & Amortization	13845	30.52	442.09	-6.14	0.03	0.18	1.05	14863.00	B
EBIT	104120	108.47	1348.85	-30329.63	0.08	0.66	5.65	60569.45	B
Part-Time Employees	3661	7.23	23.04	0.00	0.03	0.35	3.59	444.55	K
Equity Earnings/Loss (Pre-Tax, Nonrecurring)	41519	12.19	157.91	-2483.34	-0.00	0.01	0.24	7087.00	B

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	count	mean	std	min	25%	50%	75%	max	Unit
Short-Term Financial Assets	18291	203.63	2887.98	-738.25	0.01	0.17	1.93	92441.70	B
Net Financing Income/Expense	92886	-6.24	128.64	-10670.09	-0.24	-0.02	0.00	4740.00	B
Goodwill (Gross)	22899	46.16	313.86	-0.40	0.14	0.75	4.26	6958.30	B
Impairment - Financial Investments	14263	6.74	82.03	-459.28	0.00	0.01	0.15	2242.53	B
Impairment - Fixed Assets	50314	4.92	83.93	-1709.81	0.00	0.03	0.39	11072.70	B
Income Taxes	101788	28.05	340.50	-1855.99	0.01	0.12	1.23	16990.40	B
Intangible Assets (Accum. Amort. & Impair.)	34239	49.82	424.81	-52.59	0.04	0.26	1.55	18254.57	B
Intangible Assets (Gross)	34632	122.90	907.03	-60.63	0.17	1.08	6.80	37612.29	B
Intangible Assets (Net Cash Flow)	4464	18.25	193.18	-136.28	0.00	0.01	0.11	5615.31	B
Long-Term Investments	66338	105.56	1678.63	-1246.24	0.03	0.52	8.03	126930.42	B
Total Investments	85989	618.23	6893.10	-230.94	0.09	1.27	25.76	442925.68	B
Lending & Long-Term Deposits	7493	714.66	6413.50	-42.61	0.73	10.50	53.20	242431.95	B
Short-Term Loans & Receivables (Net)	40189	267.76	2113.23	-5.18	0.33	1.68	11.65	97072.45	B
Total Loans & Receivables	102232	1126.53	21694.84	-38.63	0.20	1.44	18.53	2531993.14	B
Net Cash Flow - Financing	103774	11.49	961.89	-27753.00	-0.97	-0.05	0.16	121530.63	B
Net Cash Flow - Investing	103526	-115.48	1569.11	-132477.05	-4.01	-0.40	-0.04	25880.94	B
Net Cash Flow - Operating	103874	128.39	1942.81	-80142.33	0.07	0.60	5.42	125791.99	B
Net Financial Income/Expense (Other)	17440	-1.20	29.88	-1517.46	-0.06	-0.01	-0.00	299.66	B
Net Income (After Tax)	51281	101.89	1246.50	-40408.49	0.11	0.86	6.74	44344.86	B
Nonrecurring Income/Expense	86092	-1.47	174.77	-10939.90	-0.24	-0.01	0.00	29391.89	B
Total Operating Expenses	99953	760.55	6541.08	-1143.05	0.68	4.23	31.37	227970.94	B
Total Other Assets	100956	122.21	1323.71	-41897.01	0.04	0.42	5.69	69247.18	B
Total Other Current Assets	30798	29.04	260.52	-14527.45	0.03	0.30	4.92	13300.06	B

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	count	mean	std	min	25%	50%	75%	max	Unit
Total Other Current Liabilities	89942	102.06	1122.25	-12625.25	0.04	0.40	4.97	150865.89	B
Total Other Noncurrent Liabilities	42108	115.75	2088.40	-1290.92	0.09	0.68	7.05	145414.99	B
Payables & Accrued Expenses	100432	131.24	1145.48	-0.69	0.14	0.94	8.45	77457.92	B
Plant, Machinery & Equipment (Gross)	73124	525.67	6823.00	-105.61	0.15	1.20	11.25	305445.00	B
PPE (Accum. Depreciation & Impairment)	88596	399.03	4681.38	-10168.69	0.16	1.27	11.25	227543.45	B
PPE (Gross)	92792	855.02	8895.91	-4920.35	0.46	3.59	33.30	377471.99	B
PPE (Net)	100907	437.36	4442.67	-991.57	0.20	1.84	17.45	207052.67	B
Other PPE (Gross)	65438	120.92	1761.60	-3128.77	0.03	0.31	3.70	169436.18	B
Total Provisions	89472	79.50	778.60	-166.19	0.06	0.49	4.88	44491.12	B
Long-Term Receivables & Loans	40536	100.80	1385.81	-31.92	0.01	0.14	1.48	46311.56	B
R&D Expense	32492	40.82	630.29	-136.86	0.04	0.20	1.99	22401.73	B
R&D Costs (Gross)	10642	75.39	592.23	-0.00	0.02	0.09	0.73	10374.45	B
Short-Term Debt & Notes Payable	54238	267.85	2309.18	-168.57	0.06	1.15	20.76	104779.30	B
Total Short-Term Investments	52532	131.21	1936.84	-3.40	0.03	0.33	3.48	95270.26	B
Total Assets	103976	2732.62	30953.65	0.00	2.05	12.61	137.79	1808429.76	B
Total Book Capital	103954	1154.24	10028.24	-66.81	1.29	7.70	73.41	442390.86	B
Total Current Assets	89457	505.03	4941.26	-57.13	0.50	3.03	26.51	283116.65	B
Total Current Liabilities	44789	516.29	3918.81	0.00	0.80	4.49	40.21	298411.51	B
Total Fixed Assets (Net)	82720	737.86	6722.46	-17420.67	0.59	3.86	31.22	283593.15	B
Total Liabilities	103952	2045.66	26076.03	-20.44	0.99	6.76	75.56	1700262.11	B
Total Shareholders' Equity	99822	689.10	7064.58	-11055.62	0.75	4.17	34.21	304899.93	B
Working Capital Change (Cash Flow)	98107	-26.13	623.51	-31484.37	-0.38	-0.01	0.08	35264.41	B
Avg. Receivables Collection Period (Days)	43635	95.47	919.42	-17463.43	41.87	63.48	90.74	152776.77	Unit

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	count	mean	std	min	25%	50%	75%	max	Unit
Full-Time Employees	79275	323.43	52542.63	0.00	1.68	6.93	23.05	10785002.89	K
R&D as % of Revenues	43026	287.33	8230.36	-34.17	0.68	2.58	7.06	598611.42	Unit
Turnover	62282	2.12	108.80	-17573.65	0.00	0.02	0.23	19941.57	B
Date of Incorporation	105929	1.98	0.03	1.81	1.97	1.99	2.00	2.02	K
Year	109085	2.01	0.01	2.00	2.01	2.02	2.02	2.02	K
Parent Shareholders' Equity (Total)	4123	482.85	1871.17	-63.52	6.26	55.35	326.36	37441.42	B
Business Financing Revenue (Other)	4278	45.33	542.50	-8261.00	0.00	0.04	1.33	11075.05	B
Inventories (Finished Goods)	22867	48.84	497.25	-4.83	0.03	0.22	1.57	17537.84	B
Long-Term Loans	9204	62.72	938.79	-0.00	0.01	0.19	1.91	28704.25	B
Short-Term Loans	10735	33.42	370.27	-0.00	0.01	0.22	1.96	21022.04	B
Total Noncurrent Liabilities	41426	202.81	2583.53	-99.82	0.10	0.85	6.30	131277.39	B